



(12) **United States Patent**
Elby et al.

(10) **Patent No.:** **US 12,407,439 B2**
(45) **Date of Patent:** ***Sep. 2, 2025**

(54) **SUBCARRIER BASED DATA CENTER
NETWORK ARCHITECTURE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/516,909**

(22) Filed: **Nov. 21, 2023**

(65) **Prior Publication Data**
US 2024/0283558 A1 Aug. 22, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/403,840, filed on Aug. 16, 2021, now Pat. No. 11,824,632.

(51) **Int. Cl.**
H04J 14/02 (2006.01)
H04B 10/50 (2013.01)
H04B 10/61 (2013.01)

(52) **U.S. Cl.**
CPC **H04J 14/0298** (2013.01); **H04B 10/506** (2013.01); **H04B 10/61** (2013.01)

(58) **Field of Classification Search**

CPC H04J 14/0298; H04B 10/506; H04B 10/61
See application file for complete search history.

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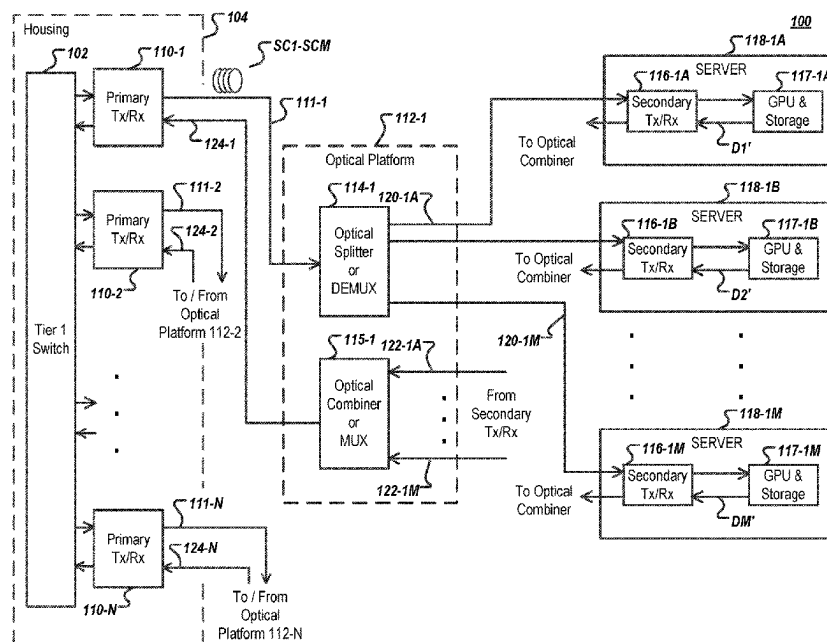
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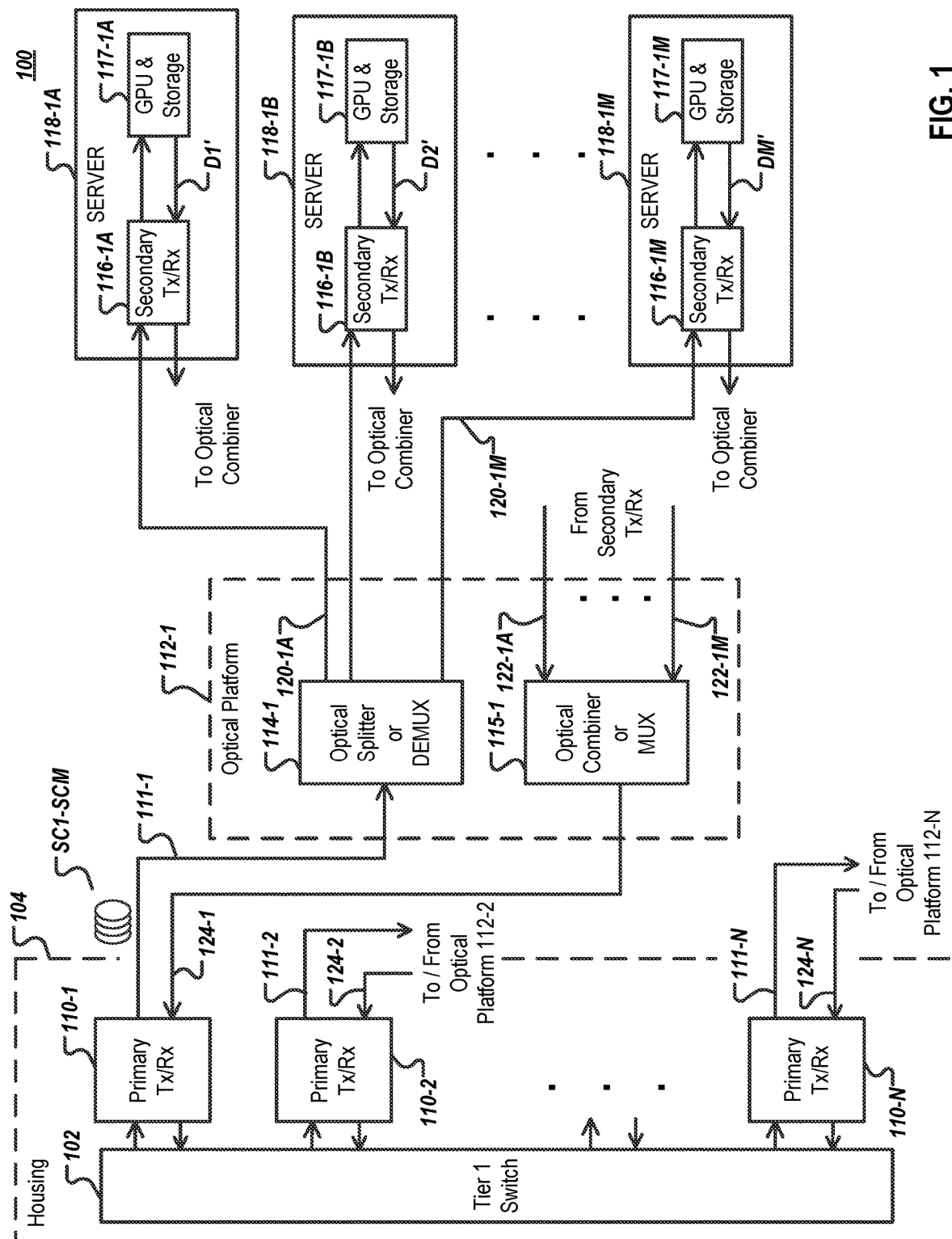
Primary Examiner — Casey L Kretzer

(57) **ABSTRACT**

Methods, systems, and apparatus, including computer programs encoded on computer storage media, for generating, transmitting, directing, receiving, and processing optical subcarriers. In some implementations, a system includes a Tier 1 switch that supplies a plurality of data channels; a transmitter that receives the plurality of data channels, the transmitter including an optical modulator that supplies a plurality of optical subcarriers based on the plurality of data channels; an optical platform that receives the plurality of optical subcarriers, the optical platform having a plurality of outputs, each of which supplying at least one of the plurality of subcarriers; a plurality of receivers, each receiving one or more of the plurality of optical subcarriers and supplying one or more of the plurality of data channels; and a plurality of servers, each of which receiving one or more of the plurality of data channels.

20 Claims, 11 Drawing Sheets





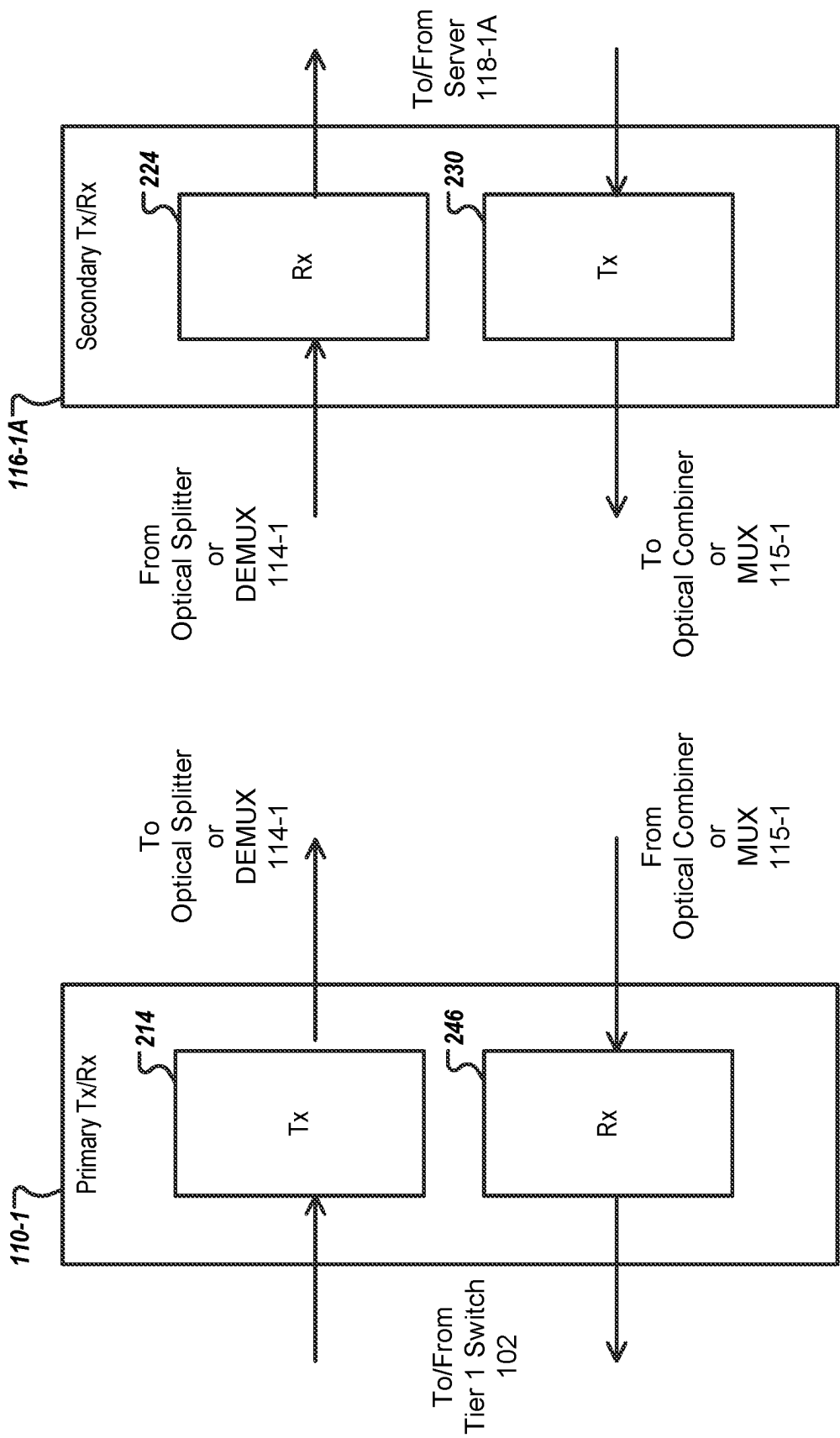


FIG. 2

300

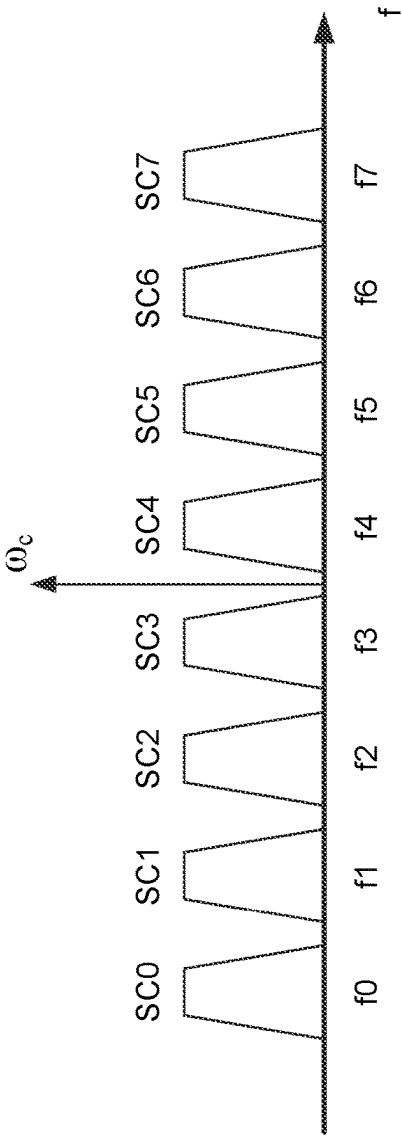


FIG. 3

214

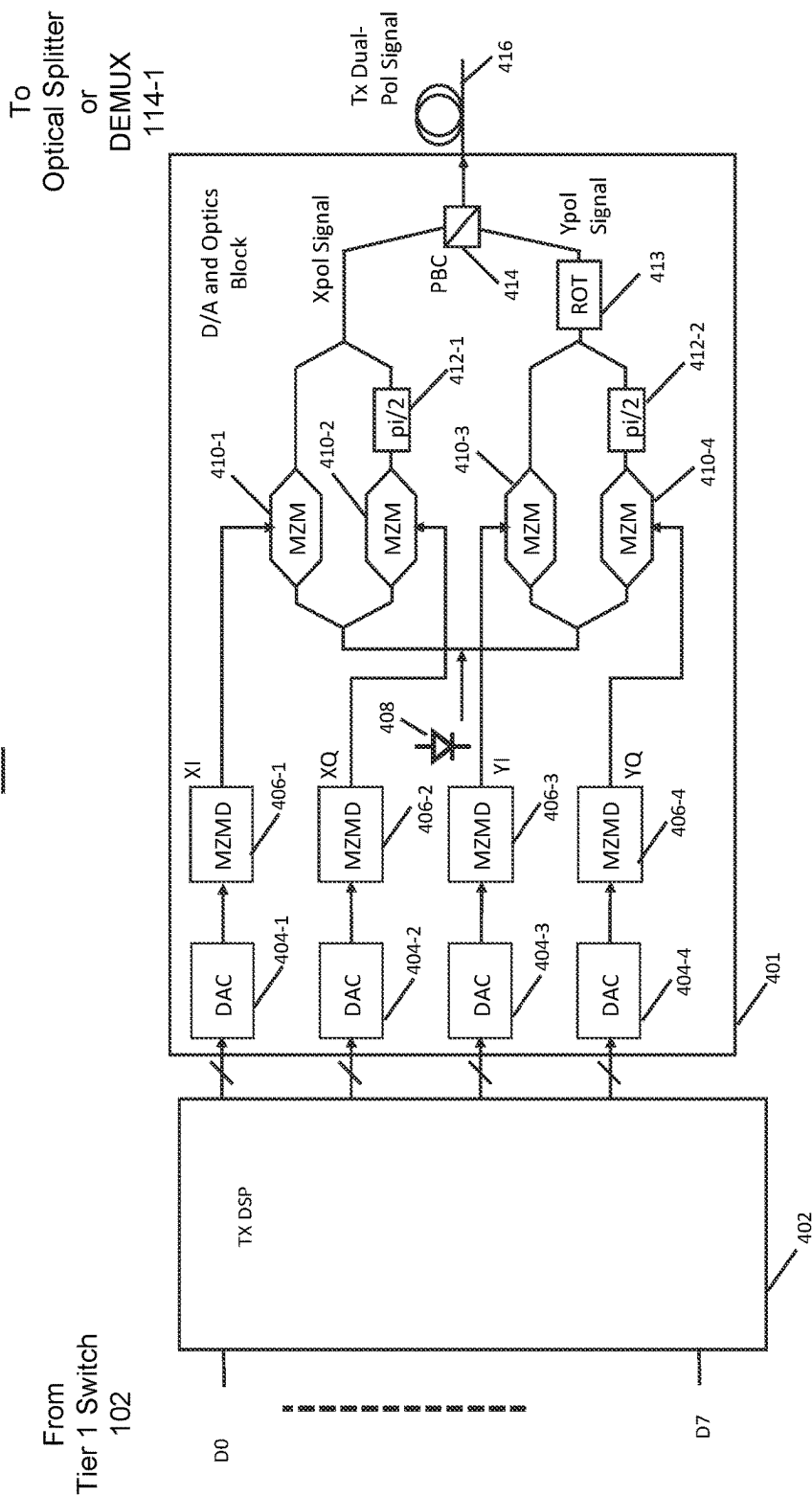


FIG. 4

402

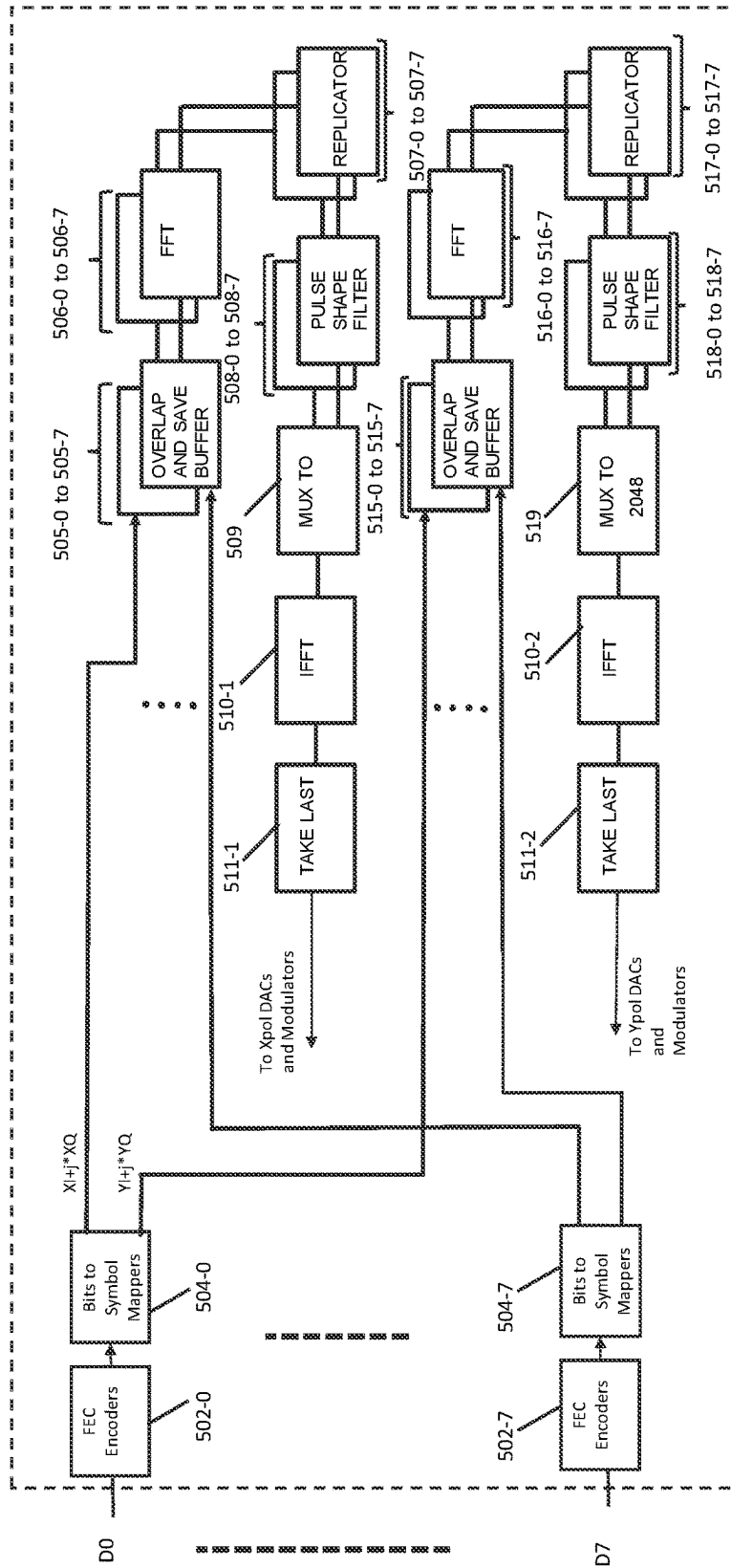


FIG. 5

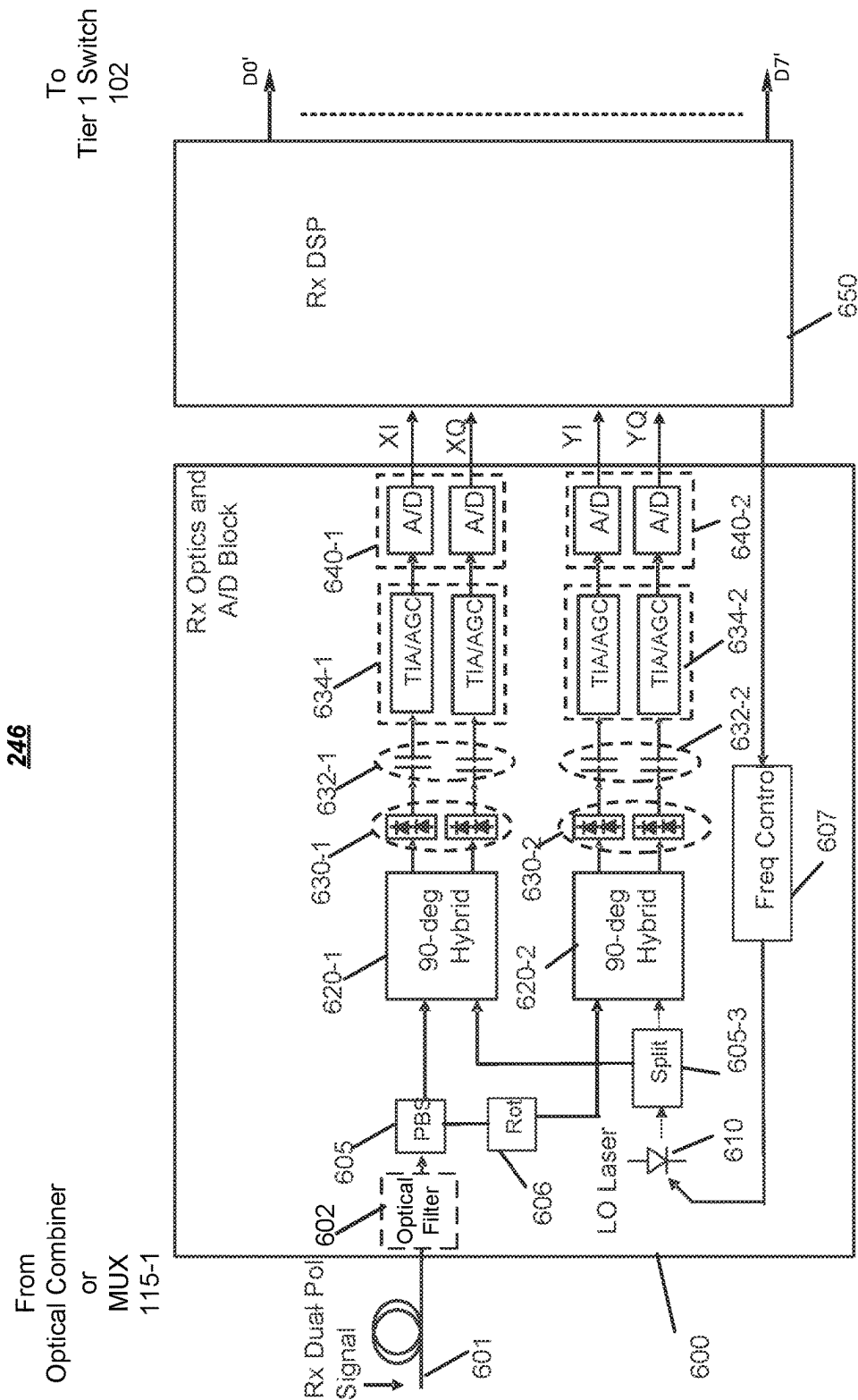


FIG. 6

650

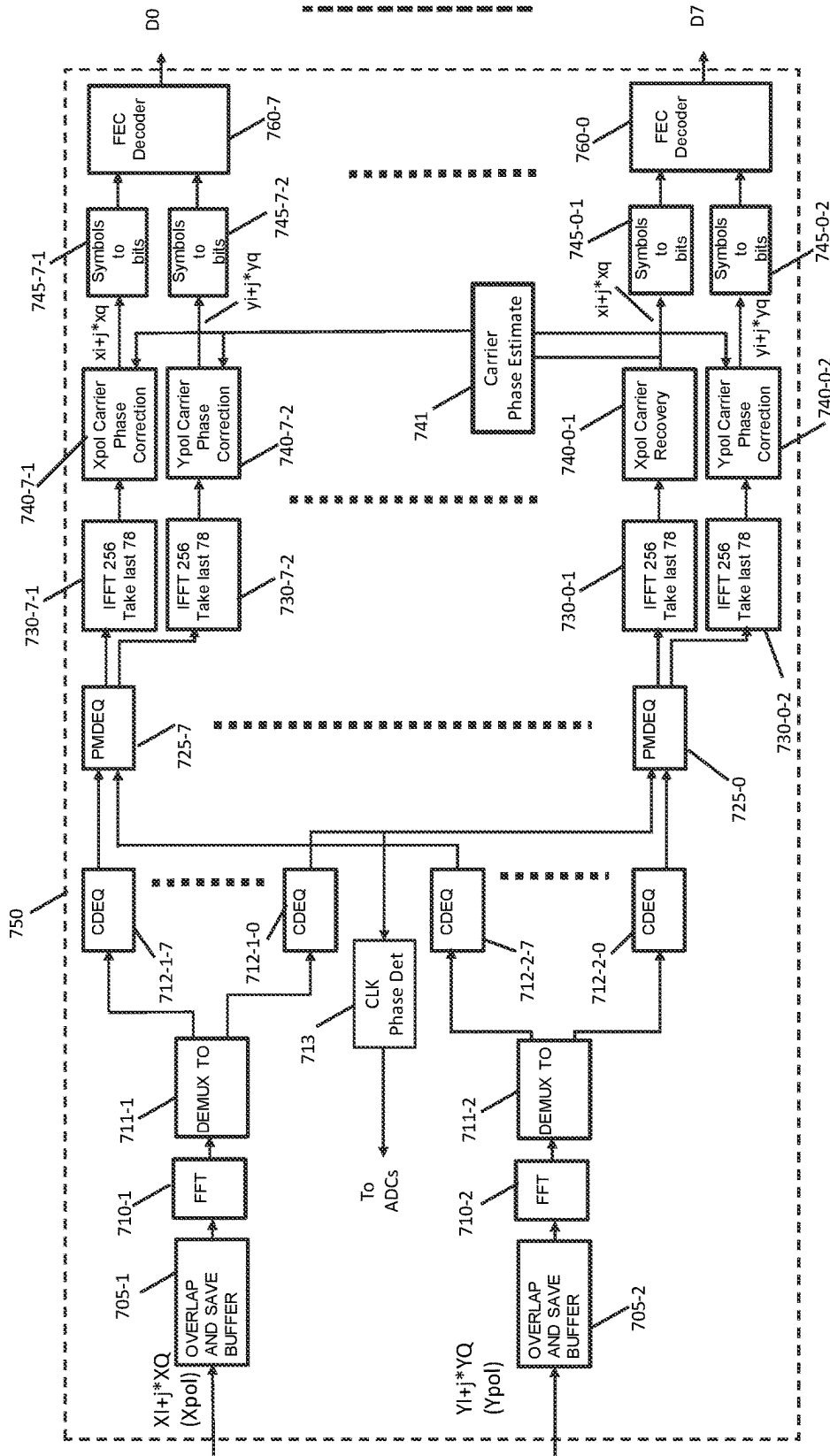


FIG. 7

230

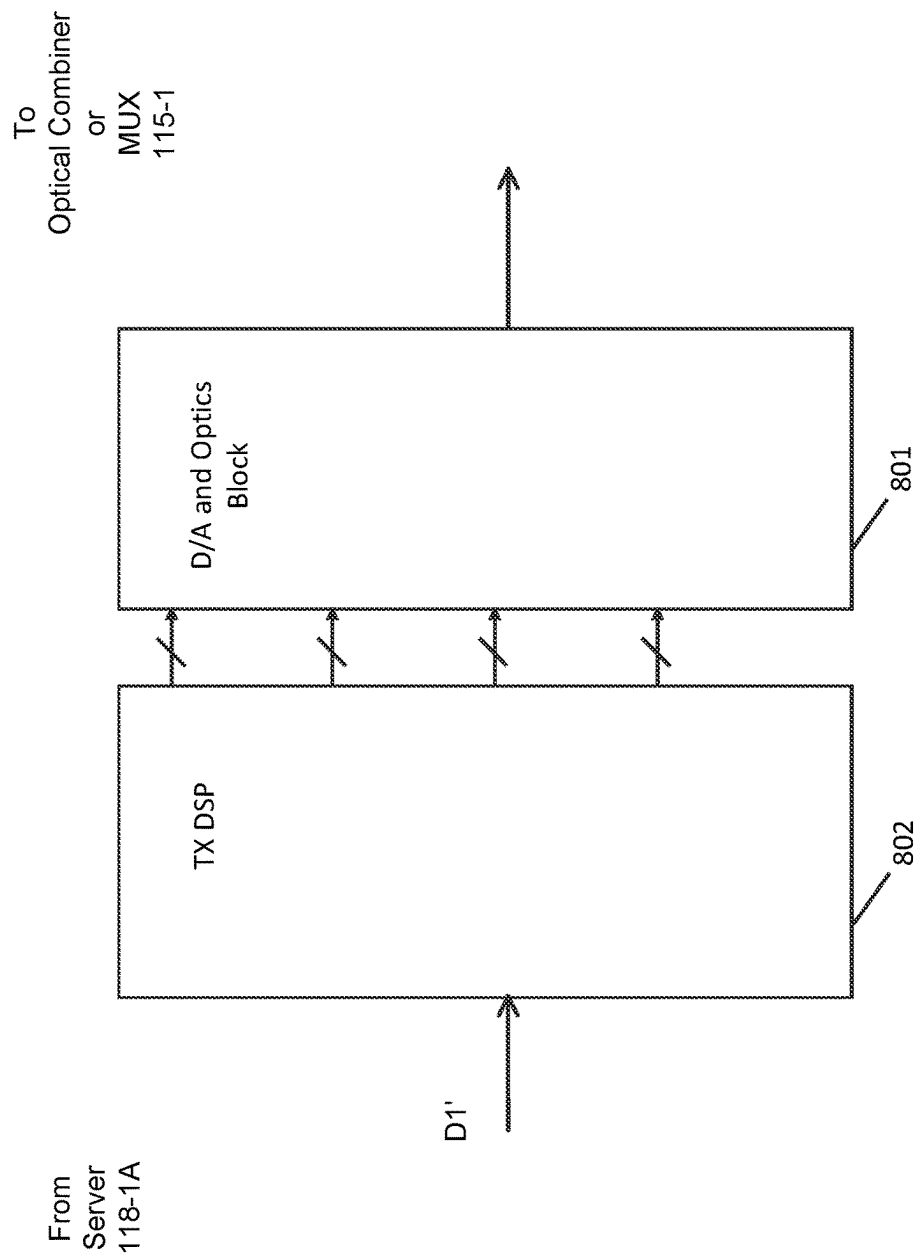


FIG. 8

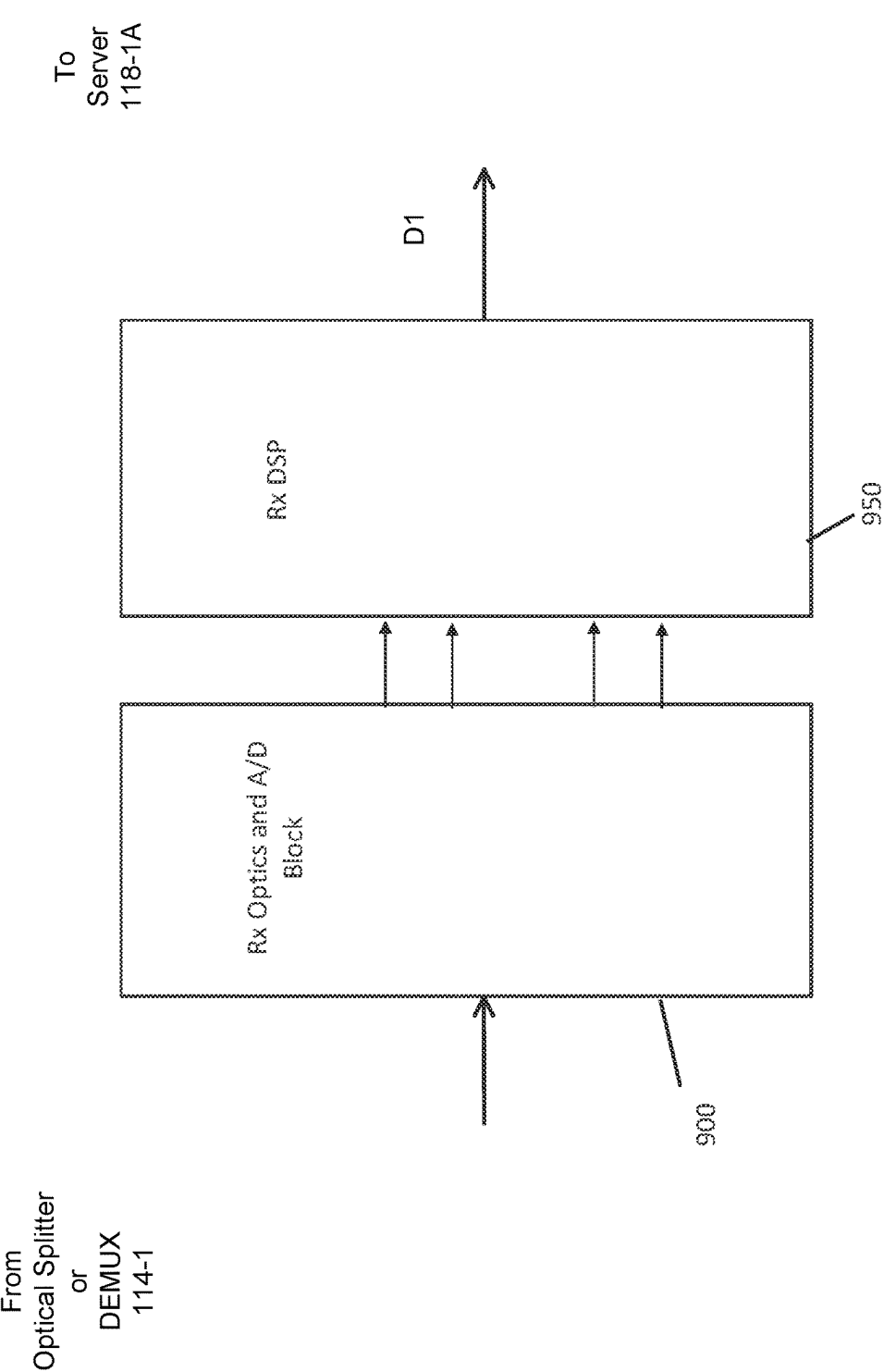
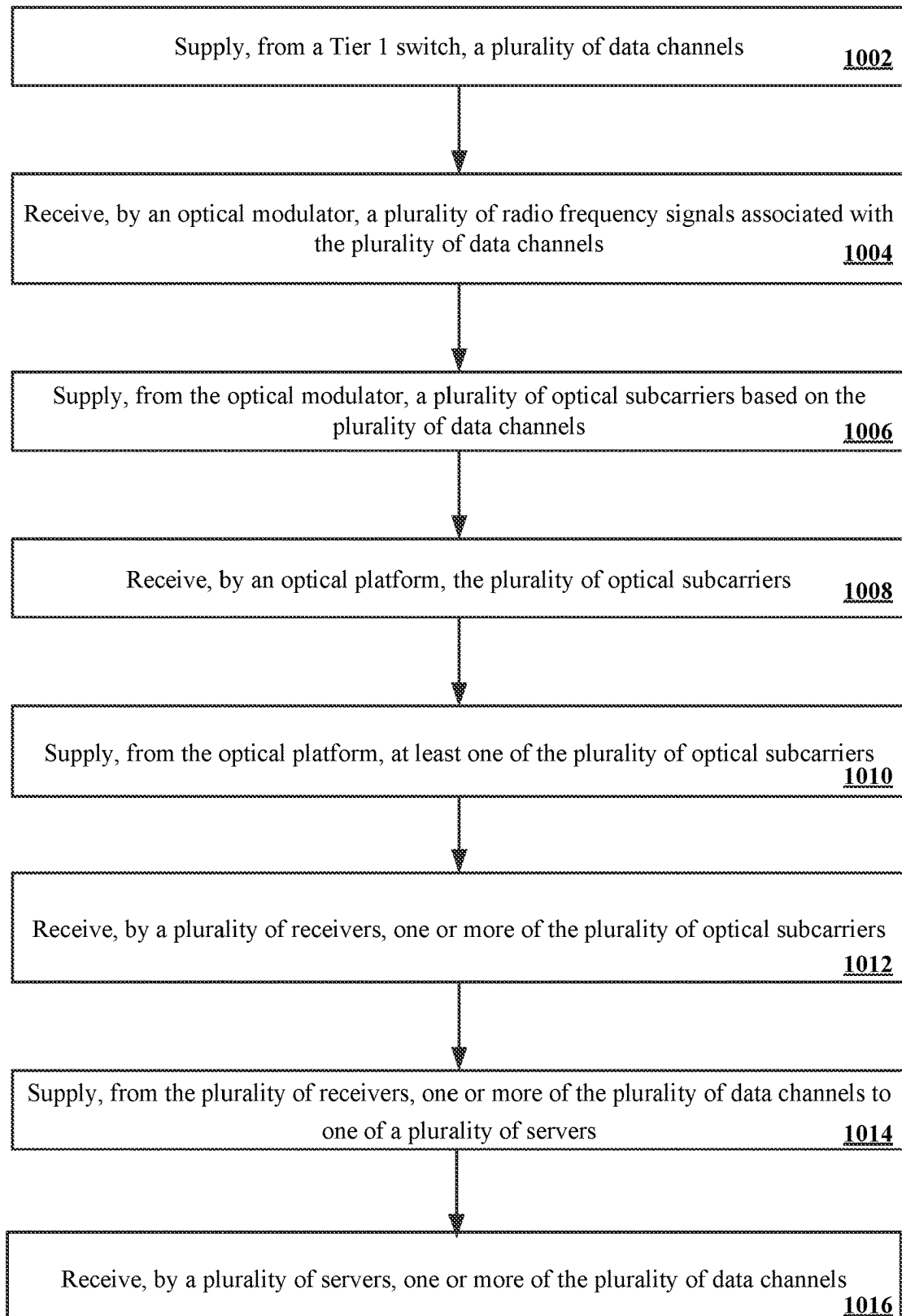
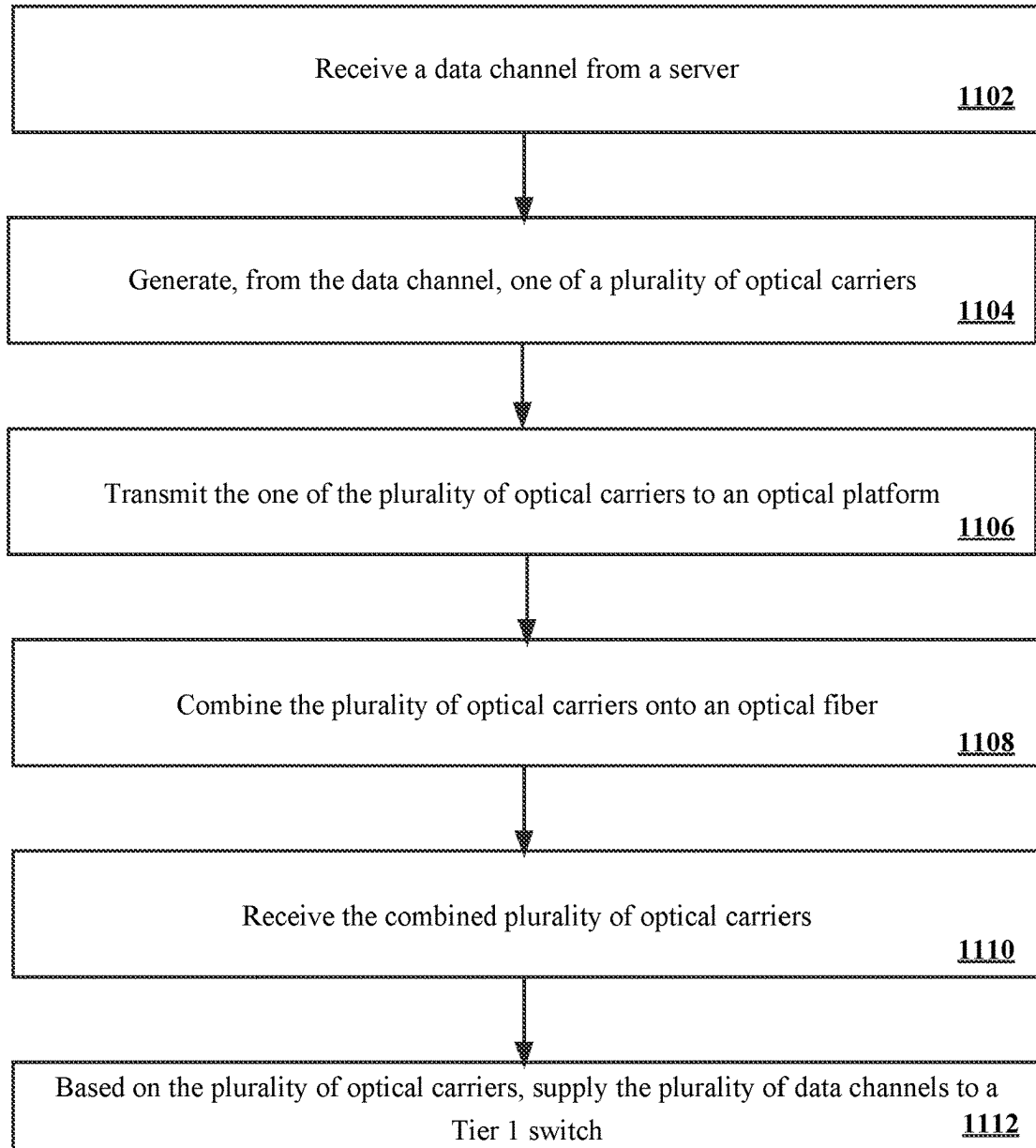


FIG. 9

1000**FIG. 10**

1100**FIG. 11**

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SUBCARRIER BASED DATA CENTER NETWORK ARCHITECTURE

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/510,528, filed Jul. 12, 2019, which claims the benefit of U.S. Provisional Patent Application No. 62/697,264, filed Jul. 12, 2018, each of which is incorporated herein by reference in its entirety.

BACKGROUND

This specification relates generally to optical communication networks.

Many optical communication networks have a leaf-spine architecture, where a series of spine switches, i.e., Tier 1 switches, each connect to a series of underlying leaf switches, i.e., top-of-rack (TOR) switches, which form an access layer for individual servers.

SUMMARY

This specification describes techniques, methods, and systems for generating, transmitting, directing, receiving, and processing optical subcarriers. In some implementations, an optical transmitter in a primary transceiver sends optical subcarriers, via an optical platform, to optical receivers, each of which being in a corresponding one of a plurality of secondary transceivers. Each optical subcarrier may be associated with data designated for a corresponding server within an optical network. In some examples, the optical subcarriers may constitute a wavelength division multiplexed (WDM) optical signal.

In general, a subcarrier-based data center network architecture, as disclosed herein, includes a primary transceiver connected to Tier 1 switch. The primary transceiver transmits a composite carrier signal that may be logically subdivided into individual optical subcarriers corresponding to the capacities of the secondary transceivers. Each of the optical subcarriers may have a corresponding one of a plurality of frequencies or wavelengths. The optical subcarriers may be transmitted to an optical platform, which, in one example, may power split the received optical subcarriers and provide a power split portion of each subcarrier to a receiver in each of the secondary transceivers. In another example, the optical platform may wavelength of frequency demultiplex the received optical subcarriers, such that each output of the optical platform supplies a respective subcarrier to a receiver in a corresponding secondary transceiver. The receiver, in turn, may demodulate one or more of the received subcarriers and supply data associated with such subcarrier(s) to a corresponding data server.

As data centers include more sophisticated servers, e.g., graphics processing unit (GPU)-based servers for virtual reality and artificial intelligence applications, higher data rates are advantageous. GPU-based servers consume significantly more power than central processing unit (CPU)-based servers. As server capacities increase, the speeds of the network interface cards (NICs) of the servers must increase commensurately. For example, NIC speeds of 100 gigabits per second (Gb/s) may be necessary for supporting complex image and visual processing. As NIC speeds increase, the Tier 1 switch speeds must also increase, as each Tier 1 link may carry traffic from multiple NICs. For example, each Tier 1 link may carry traffic from 8, 16, 32, or 64 NICs.

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A system with a typical leaf-spine architecture includes multiple servers that communicate with top-of-rack (TOR) switches, i.e. leaf switches, which in turn communicate with Tier 1 switches, i.e., spine switches. In a typical leaf-spine architecture, the TOR processes data packets to enable traffic aggregation and distribution in the upstream direction towards the Tier 1 switches, and to provide correct forwarding of the data packets to the appropriate NIC in the downstream direction towards the servers. TOR packet switches, however, consume significant power, and the power consumption scales up as port speeds scale up.

The system and techniques described herein can be used to replace the TOR within an optical communications system with an optical platform, including an optical splitter and an optical power combiner. Subcarrier based data center network architecture using an optical platform consumes less power, compared to a TOR, while at the same time enable higher data rates to be directed to the servers. Thus, the techniques disclosed herein may reduce the power and space required for optical communications systems, while enabling higher data rates.

In general, one innovative aspect of the subject matter described in this specification can be embodied in a system that includes a Tier 1 switch that supplies a plurality of data channels; a transmitter that receives the plurality of data channels, the transmitter including a laser, and an optical modulator that receives a plurality of radio frequency (RF) signals associated with the plurality of data channels, such that the optical modulator supplies a plurality of optical subcarriers based on the plurality of data channels; an optical platform, including one of an optical demultiplexer or a splitter, that receives the plurality of optical subcarriers, the optical platform having a plurality of outputs, each of which supplying at least one of the plurality of subcarriers; a plurality of receivers, each of which being coupled to a respective one of the plurality of outputs of the optical platform, each of the plurality of receivers receiving one or more of the plurality of optical subcarriers and supplying one or more of the plurality of data channels based on the received one or more of the plurality subcarriers; and a plurality of servers, each of which receiving one or more of the plurality of data channels from one or more of the plurality of receivers.

In general, one innovative aspect of the subject matter described in this specification can be embodied in a system that includes a plurality of servers, each of which supplying a corresponding one of a plurality of data channels; a plurality of transmitters, each of which receiving a corresponding one of the plurality of data channels, each of the plurality of transmitters providing a respective one of a plurality of optical carriers, each of the plurality of optical carriers being modulated in accordance with a respective one of a plurality of radio frequency (RF) signals, each RF signal being based on a respective one of the plurality of data channels; an optical platform that combines the plurality of optical carriers onto an optical fiber; a receiver that receives the combined plurality of optical carriers, the receiver including a local oscillator, such that based on plurality of optical carriers and an output of the local oscillator, the receiver supplies the plurality of data channels; and a Tier 1 switch that receives the plurality of data channels.

Particular implementations may include one or more of the following features, alone or in combination. The optical platform may include an optical demultiplexer that supplies a corresponding one or more of the plurality of subcarriers at each of the plurality of outputs. The optical platform may include an optical splitter that supplies a copy of the plurality

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of subcarriers at each of the plurality of outputs. Each of the plurality of optical subcarriers may be a Nyquist subcarrier. Each of the plurality of data channels supplied to the transmitter may constitute a respective one of a plurality of electrical signals. The Tier 1 switch may include a housing, the housing including the transmitter. Each of the plurality of optical subcarriers may be modulated in accordance with a modulation format, the modulation format being selected from an m-quadrature amplitude modulation (QAM), m being an integer, quadrature phase shift keying (QPSK), and binary phase shift keying (BPSK). Each of the plurality of receivers may include a respective one of a plurality of local oscillator lasers. One of the plurality of receivers may include a local oscillator laser; a control circuit, the control circuit being coupled to the local oscillator laser to thereby control a frequency of light output from the local oscillator laser; a photodetector circuit configured to receive at least a portion of the light output from the local oscillator laser and at least part of one of the plurality of outputs of the optical platform; and a processor circuit that supplies one of the plurality of data channels based on an output of the photodetector circuit. The processor may supply said one of the plurality of data channels further based on a frequency of the light output from the local oscillator. Each of the plurality of receivers includes a corresponding one of a plurality of optical filters, each of the plurality of optical filters being configured to select a respective one of the plurality of optical subcarriers. The processor circuit includes a filter circuit, the processor circuit supplying said one of the plurality of data channels based on an output of the filter circuit.

In some implementations, the optical platform may include an optical power combiner. The optical platform may include an optical multiplexer. The optical platform may include an arrayed waveguide grating. The plurality of optical carriers may be included in a group of carriers provided to the receiver, the receiver including an optical filter that selects the plurality of optical carriers; and a photodetector circuit that converts the plurality of optical carriers to electrical signals. The local oscillator may be an electrical local oscillator, the plurality of data channels being output further based on an output of the electrical local oscillator and the electrical signals. Each of the plurality of optical carriers may be a Nyquist carrier. Each of the plurality of data channels may constitute a respective one of a plurality of electrical signals. The Tier 1 switch may include a housing, the housing including the receiver. Each of the plurality of optical carriers may be modulated in accordance with a modulation format, the modulation format being selected from an m-quadrature amplitude modulation (QAM), m being an integer, quadrature phase shift keying (QPSK), and binary phase shift keying (BPSK). The local oscillator may include a local oscillator laser.

Other embodiments of this aspect include corresponding computer systems, apparatus, computer program products, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the methods. A system of one or more computers can be configured to perform particular operations or actions by virtue of having software, firmware, hardware, or a combination of them installed on the system that in operation causes or cause the system to perform the actions. One or more computer programs can be configured to perform particular operations or actions by virtue of including instructions that, when executed by data processing apparatus, cause the apparatus to perform the actions.

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Additional embodiments of this aspect include corresponding communications systems, apparatus, switches, and nodes, each configured to perform the actions of the methods. For example, a communications system of one or more switches can be configured to perform particular operations including processing and routing data traffic upstream and downstream within the communications system.

The details of one or more embodiments of the subject matter of this specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram that illustrates an example of a system for subcarrier based optical communication.

FIG. 2 is a block diagram that illustrates example primary transceivers and secondary transceivers.

FIG. 3 illustrates an example of a composite carrier signal.

FIG. 4 illustrates an example of a transmitter of a primary transceiver.

FIG. 5 illustrates an example of a transmitter digital signal processor.

FIG. 6 illustrates an example of a receiver of a primary transceiver.

FIG. 7 illustrates an example of a receiver digital signal processor.

FIG. 8 illustrates an example of a transmitter of a secondary transceiver.

FIG. 9 illustrates an example of a receiver of a secondary transceiver.

FIG. 10 is a flow diagram that illustrates an example of a process for transmitting an optical signal downstream from a Tier 1 switch to a server.

FIG. 11 is a flow diagram that illustrates an example of a process for transmitting an optical signal upstream from a server to a Tier 1 switch.

Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

Consistent with the present disclosure, a Tier 1 Switch supplies data to a high capacity transmitter that transmits a plurality of optical subcarriers to a passive optical platform. The optical platform may include a passive optical splitter that supplies a power split portion of the optical subcarriers to lower rate receivers. Each subcarrier may carry information indicative of data for output to a particular server. Accordingly, in one example, each receiver is coupled to a corresponding server and demodulates a designated one of the received subcarriers to thereby output data intended for such server. In another example, the optical platform wavelength or frequency demultiplexes the optical subcarriers received from the high capacity transmitter and supplies each subcarrier to a corresponding receiver, which, in a similar fashion demodulates the received subcarrier and supplies the resulting data to a respective server.

The passive optical platform may consume little or no power and, is therefore, less expensive to operate than a conventional TOR and does not require the cooling mechanisms that a TOR would otherwise require. Moreover, as described in greater detail below, optical subcarriers may be readily added by modulating the output of a laser in the high speed transmitter without adding additional lasers, for

example. Accordingly, a system consistent with the present disclosure may be easily expanded.

Reference will now be made in detail to the present embodiments of the present disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

FIG. 1 is a block diagram that illustrates an example of a system 100 for subcarrier based optical communication. The system 100 includes a Tier 1 switch 102, a plurality of primary transceivers, a plurality of optical platforms, a plurality of secondary transceivers, and a plurality of servers. For example, the system 100 may include N number of primary transceivers 110-1 to 110-N, N number of optical platforms numbered 112-1 to 112-N, N×M number of secondary transceivers 116-1A to 116-NM, and N×M number of servers 118-1A to 118-NM, where M number of servers communicates with each of the N number of primary transceivers.

Communication through the system 100 can occur in both the downstream and upstream directions. In the downstream direction, data flows from the Tier 1 switch 102 to the servers 118-1A to 118-NM. In the upstream direction, data flows from the servers 118-1A to 118-NM to the Tier 1 switch 102.

The Tier 1 switch 102 can be, for example, a Tier 1 switch within an optical network. The Tier 1 switch 102 may include one or more routing modules, one or more uplink ports, and one or more input/output (I/O) ports. For example, the Tier 1 switch 102 may include N=8 I/O ports. The Tier 1 switch 102 can operate at high speeds, for example, 800 Gb/s.

The Tier 1 switch 102 manages the flow of data across a network, e.g., the system 100. In the downstream direction, the Tier 1 switch 102 may receive incoming data, e.g., a plurality of data channels D1 to DM, and direct or designate the data to a destination server. The plurality of data channels D1 to DM may constitute, for example, a plurality of electrical signals. In some examples, the Tier 1 switch 102 may receive incoming data from one or more network core switches through an uplink port, e.g., an Ethernet port. The Tier 1 switch 102 may examine the destination address of the incoming data and compare the address to a table of network addresses corresponding to the NICs of the servers within the optical network. The network addresses can be, for example, Internet Protocol (IP) addresses.

The Tier 1 switch 102 communicates with N number of primary transceivers numbered 110-1 to 110-N. The primary transceivers 110-1 to 110-N can each connect to the Tier 1 switch. In some examples, the Tier 1 switch 102 includes a housing 104, and the housing 104 houses the primary transceiver 110-1. The Tier 1 switch 102 dispatches the data to the appropriate primary transceivers 110-1 to 110-N.

A transmitter (described below) provided in each of the primary transceivers 110-1 to 110-N may generate a composite carrier signal 111-1 to 111-N at large bandwidths of M times the bandwidth of each of the servers 118-1A to 118-1M. As further described below, each composite carrier signal 111-1 to 111-N may include a plurality of optical subcarriers SC1 to SCM. For example, if the primary transceiver 110-1 communicates with M=8 servers, each server having a bandwidth of 100 Gigahertz (GHz), the primary transceiver 110-1 generates a composite carrier signal 111-1 at a bandwidth of 800 GHz. As a result of encoding by the transmitter, the information carried by each subcarrier SC1 to SCM is representative of a corresponding data channel D1 to DM.

The composite carrier signal 111-1 includes a plurality of optical subcarriers SC1 to SCM. Each subcarrier SC1 to SCM can carry information independently of the other subcarriers. Each subcarrier SC1 to SCM may have a bandwidth of 1/M times the bandwidth of the composite carrier signal 111-1. For example, the composite carrier signal 111-1 may have a bandwidth of 800 GHz, and may transmit to M=8 servers. Each subcarrier SC1-SC8 within the composite carrier signal 111-1 may therefore have a bandwidth of 1/8 times 800 GHz, or 100 GHz.

The primary transceivers 110-1 to 110-N can operate at M times the speed of the secondary transceivers 116-1A to 116-NM. For example, M=8 secondary transceivers 116-1A to 116-1H may each operate at a speed of 100 Gb/s. The primary transceiver 110-1 therefore may operate at a speed of 8 times 100 Gb/s, or 800 Gb/s.

The system 100 includes N number of optical platforms numbered 112-1 to 112-N. Each primary transceiver 110-1 to 110-N communicates with an optical platform 112-1 to 112-N via one or more optical fibers. For example, the primary transceiver 110-1 may communicate with the optical platform 112-1, and the primary transceiver 110-2 may communicate with the optical platform 112-2.

The primary transceiver 110-1 outputs the composite carrier signal 111-1, including the plurality of optical subcarriers SC1 to SCM, onto an optical fiber. The composite carrier signal 111-1 transmits via the optical fiber to the optical platform 112-1.

The optical platform 112-1 receives the composite carrier signal 111-1, including the plurality of subcarriers SC1 to SCM, from the primary transceiver 110-1. The optical platform 112-1 may include one of an optical splitter or an optical demultiplexer (DEMUX) 114-1. The optical platform 112-1 also may include one of an optical power combiner or an optical power multiplexer (MUX) 115-1.

In the downstream direction, i.e., from the Tier 1 switch 102 to the servers 118-1A to 118-1M, the optical splitter 114-1 of the optical platform 112-1, for example, has an that receives composite signal 111-1 including the plurality of subcarriers SC1 to SCM from a transmitter in transceiver 110-1. For example, the optical splitter may split the composite carrier signal 111-1 into multiple power split portions (120-1A to 120-1M), each of which being provided at a respective output of the splitter 114-1. Accordingly, a power split portion or copy of each subcarrier is output from splitter 114-1. In some examples, each of the outputs 120-1A to 120-1M may have a power lower than the power of the composite carrier signal 111-1. For example, the power of each of the outputs may be 1/M times the power of the composite carrier signal 111-1. The optical splitter 114-1 outputs the same subcarriers that are input to the optical splitter 114-1, such that each of the outputs 120-1A to 120-1M may include each of the subcarriers SC1 to SCM. In another example, one or more optical amplifiers may be provided to increase the power of outputs of splitter 114-1.

In some examples, the optical platform 112-1 may include an optical demultiplexer instead of the optical splitter. The optical demultiplexer may supply one or more of the plurality of subcarriers SC1 to SCM at each of the plurality of outputs. For example, each of the outputs of the demultiplexer may include a group of subcarriers selected from the subcarriers SC1 to SCM, or may include a corresponding one of the subcarriers SC1 to SCM. The optical demultiplexer may include a wavelength selective switch (WSS), a reconfigurable optical add-drop multiplexer (ROADM), or a passive demultiplexer. For example, each of the plurality of

subcarriers may be spectrally spaced to allow a WSS to select specific subcarriers for supplying at specific outputs.

Each optical platform **112-1** to **112-N** may communicate with M number of secondary transceivers. For example, optical platform **112-1** may communicate with secondary transceivers **116-1A** to **116-1M**, and optical platform **112-2** may communicate with secondary transceivers **116-2A** to **116-2M**.

The optical platform **112-1** to **112-N** can be used in a leaf-spine architecture as a replacement for a TOR and, therefore, consume less power than a convention systems, as noted above.

In the downstream direction, the optical platform **112-1** directs, to a plurality of receivers, each within a secondary transceiver **116-1A** to **116-1M**, one of the outputs **120-1A** to **120-1M** via optical fibers. A receiver in each secondary transceivers **116-1A** to **116-1M** each may filter, or isolate, the single subcarrier SC1 to SCM associated with a corresponding server **118-1A** to **118-1M**.

Each secondary transceiver **116-1A** to **116-1M** connects with a corresponding one of servers **118-1A** to **118-1M**. For example, the secondary transceiver **116-1A** may connect with the server **118-1A**, and the secondary transceiver **116-1B** may connect with the server **118-1B**. The secondary transceivers **116-1A** to **116-1M** output the data from a respective one of outputs **120-1A** to **120-1M** to a corresponding one of the plurality of servers **118-1A** to **118-1M**.

In some implementations, the server **118-1A** may include the secondary transceiver **116-1A**, and a GPU and storage module **117-1A**. For example, the server **118-1A** may include a housing, where the housing houses the secondary transceiver **116-1A** and/or the GPU and storage module **117-1A**. The storage may hold data and instructions that are entered through the secondary transceiver **116-1A**, before they are processed by the GPU. The storage may also save the data for later use. The GPU includes a control unit and an arithmetic logic unit (ALU), and directs the activities of the server **118-1A**. In the downstream direction, the GPU processes the data from the subcarrier associated with the server **118-1A**, and carries out instructions to the server **118-1A**.

In the upstream direction, i.e., from the servers **118-1A** to **118-1M** to the Tier 1 switch **102**, the GPU generates data and may send data to the storage. A transmitter in the secondary transceiver **116-1A** may receive the data from the storage and process the data for transmission to upstream components of the optical network. The data may include, for example, a corresponding one of a plurality of data channels D1' to DM'. The plurality of data channels D1' to DM' may constitute, for example, a plurality of electrical signals. Each transmitter in the secondary transceivers **116** may output, in one example, an optical signal, such as an optical subcarrier or carrier having a frequency that is the same as or similar to that of the received subcarrier. Moreover, each upstream optical subcarrier is associated with a corresponding one of data channels D1' to DM'.

In the upstream direction, the optical power combiner **115-1** of the optical platform **112-1** can receive carrier signals, e.g., the carrier signals **122-1A** to **122-1M**, from a plurality of transmitters, each within a secondary transceiver **116-1A** to **116-1M**. The optical power combiner **115-1** may combine the multiple carrier signals **122-1A** to **122-1M** into a combined plurality of optical carriers **124-1** for forwarding to the Tier 1 switch **102**.

In some examples, the optical platform **112-1** may include an optical multiplexer instead of an optical combiner. The optical multiplexer may receive one or more of a plurality of

carriers at each of a plurality of inputs. For example, each of the inputs to the multiplexer may include a group of carriers or a single carrier. The optical multiplexer may include, for example, an arrayed waveguide grating. The arrayed waveguide grating may include waveguides with various path lengths that cause constructive interference between the carrier signals **122-1A** to **122-1M**. Thus, the arrayed waveguide grating may be used to combine the carrier signals **122-1A** to **122-1M** onto an optical fiber.

While FIG. 1 shows the system **100** as including a particular quantity and arrangement of components, in some implementations, the system **100** may include additional components, fewer components, different components, or differently arranged components. For example, the quantity of primary transceivers **110-1** to **110-N**, and secondary transceivers **116-1A** to **116-1M**, may vary based on the requirements of the system **100**. In some instances, each primary transceiver may communicate with a different number of servers. For example, primary transceiver **110-1** may communicate with M number of servers, while primary transceiver **110-2** may communicate with P number of servers. In some instances, one of the components illustrated in FIG. 1 may carry out a function described herein as being carried out by another one of the components illustrated in FIG. 1.

FIG. 2 is a block diagram that illustrates an example primary transceiver and secondary transceiver. For example, FIG. 2 illustrates a more detailed depiction of the primary transceiver **110-1**, and the secondary transceiver **116-1**, from FIG. 1. As shown in FIG. 1, a system, e.g., the system **100**, may include N number of primary transceivers and NxM number of secondary transceivers. Each of the N number of primary transceivers may include a transmitter **214** and a receiver **246**. Each of the NxM number of secondary transceivers may include a transmitter **230** and a receiver **224**.

In some examples, the primary transceiver **110-1** is an external device that plugs into the I/O ports of the Tier 1 switch **102**. For example, the primary transceiver **110-1** can be packaged in industry standard pluggable form factors, e.g., small form-factor pluggable (SFP) or C form-factor pluggable (CFP) transceivers.

In some examples, the primary transceiver **110-1** is built into the Tier 1 switch **102**. For example, the primary transceiver **110-1** can be integrated directly with the motherboard or daughter card of the Tier 1 switch **102**. In some examples, the Tier 1 switch **102** includes a housing, e.g., the housing **104**, and the housing **104** houses the primary transceiver **110-1**.

The primary transceiver **110-1** includes a transmitter **214** and a receiver **246**. In the downstream direction, the transmitter **214** receives data channels through the I/O port of the Tier 1 switch **102**. The transmitter **214** multiplexes, or combines, the data into a multiplexed optical signal **111-1** for transmission to the destination servers via the passive optical splitter **114-1** of the optical platform **112-1**.

In the upstream direction, the receiver **246** of the primary transceiver **110-1** may receive the combined plurality of optical carriers **124-1** from the optical power combiner **115-1** of the optical platform **112-1**. The receiver **246** outputs the data channels to the Tier 1 switch **102** through an I/O port. Receiver **246** is described in greater detail below.

In some examples, the secondary transceiver **116-1A** is an external device that plugs into the server **118-1A**. For example, the secondary transceiver **116-1A** can be packaged in industry standard pluggable form factor, e.g., a small form-factor pluggable (SFP) or C form-factor pluggable (CFP) transceiver.

In some examples, the secondary transceiver **116-1A** is built into the server **118-1A**. For example, the secondary transceiver **116-1A** can be integrated directly with the motherboard or daughter card of the server **118-1A**. In some examples, the server **118-1A** includes a housing, and the housing houses the secondary transceiver **116-1A**.

The secondary transceiver **116-1A** includes a transmitter **230** and a receiver **224**. Each of the receivers **224** within each of the secondary transceivers **116-1A** to **116-1M** is coupled to a respective one of the plurality of outputs **120-1A** to **120-1M** of the optical platform **112-1**.

In the downstream direction, the secondary transceiver **116-1A** receives the output **120-1A** from the splitter **114-1** of the optical platform **112-1**. The receiver **224** receives the output **120-1A**, which includes the plurality of optical subcarriers **SC1** to **SCM**, and isolates the specific subcarrier of the output **120-1A** that is associated with the server **118-1A**. The receiver **224** of the secondary transceiver **116-1A** demodulates and decodes the output **120-1A** into a digital data format for processing by the server **118-1A**.

The receiver **224** supplies to a corresponding server one or more of the plurality of data channels **D1** to **DM** based on the received plurality of subcarriers **SC1** to **SCM**. For example, the receiver **224** may demodulate and process optical subcarrier **SC1** to supply the data channel **D1** associated with subcarrier **SC1** to the server **118-1A**. Each of the servers **118-1A** to **118-1M** receives one or more of the plurality of data channels from one or more of the plurality of receivers. For example, the server **118-1A** may receive the data channel **D1** from the receiver **224** of the transceiver **116-1A**.

In the upstream direction, a plurality of servers supplies a corresponding one of a plurality of data channels. Each of the plurality of transmitters receives a corresponding one of the plurality of data channels. For example, the transmitter **230** of the secondary transceiver **116-1A** receives the data **D1'** from the server **118-1A** and transmits a subcarrier or carrier that carries information indicative of data **D1'**, via an optical fiber, to combiner **115-1**, which combines the received carrier from transmitter **230** with carriers or subcarriers carrying a respective one of data **D2'** to **DM'** output from transmitters of the other secondary transceivers **116** and supplies such combined carriers or subcarriers to receiver **246** of primary node **110-1**.

FIG. 3 illustrates an example of a composite carrier signal **300**. The composite carrier signal may be, for example, the composite carrier signal **111-1** and/or the outputs **120-1A** to **120-1M**. The composite carrier signal **300** includes $M=8$ optical subcarriers **SC0** to **SC7**. The composite carrier signal **300** is graphed in the frequency spectrum. The subcarriers may be, for example, Nyquist subcarriers. The center frequency of each of subcarriers **SC0** to **SC7** is f_0 to f_7 , respectively, such that f_0 is the center frequency of **SC0**, f_1 is the center frequency of **SC1**, etc. The center frequency of the composite carrier signal is $@c$, which corresponds to the frequency of light output from a laser in transmitter **214** prior to modulation. Each subcarrier may carry information indicative of data for output to a particular server. For example, subcarrier **SC0** may carry information indicative of data channel **DO**.

FIG. 4 illustrates an example of a transmitter **214** of a primary transceiver **110-1** consistent with the present disclosure. The transmitter **214** in this example is configured to provide data to $M=8$ servers by a modulated optical signal with 8 subcarriers. The transmitter **214** may include a digital signal processor (DSP) **402**, which, in this example, receives a plurality of data inputs **D0-D7**. It is understood, however,

that DSP **402** may receive more or fewer inputs. Based on data inputs **D0-D7**, DSP **402** may supply a plurality of outputs to D/A and optics block **401** including digital-to-analog conversion (DAC) circuits **404-1** to **404-4**, which convert digital signal received from DSP **402** into corresponding analog radio frequency (RF) signals. D/A and optics block **401** may also include driver circuits **406-1** to **406-2** that receive the plurality of RF signals from DACs **404-1** to **404-4** and adjust the voltages or other characteristics thereof to provide drive signals to a corresponding one of optical modulators **410-1** to **410-4**.

D/A and optics block **401** further includes optical modulators **410-1** to **410-4**, each of which may be a Mach-Zehnder modulator (MZM) that modulates the phase and/or amplitude of the light output from laser **408**. As further shown in FIG. 4, light output from laser **408**, also included in block **401** is split, such that a first portion of the light is supplied to a first MZM pairing including MZMs **410-1** and **410-2**, and a second portion of the light is supplied to a second MZM pairing including MZMs **410-3** and **410-4**. The first portion of the light is further split into third and fourth portions, such that the third portion is modulated by MZM **410-1** to provide an in-phase (I) component of an X (or TE) polarization component of a modulated optical signal, and the fourth portion is modulated by MZM **410-2** and fed to phase shifter **412-1** to shift the phase of such light by 90 degrees in order to provide a quadrature (Q) component of an X polarization component of the modulated optical signal. Similarly, the second portion of the light is further split into fifth and sixth portions, such that the fifth portion is modulated by MZM **410-3** to provide an I component of a Y (or TM) polarization component of the modulated optical signal, and the sixth portion is modulated by MZM **410-4** and fed to phase shifter **412-2** to shift the phase of such light by 90 degrees to provide a Q component of a Y polarization component of the modulated optical signal.

The optical outputs of MZMs **410-1** and **410-2** are combined to provide an X polarized optical signal including I and Q components and fed to a polarization beam combiner (PBC) **414** provided in block **401**. In addition, the outputs of MZMs **410-3** and **410-4** are combined to provide an optical signal that is fed to polarization rotator **413**, further provided in block **401**, which rotates the polarization of such optical signal to provide a modulated optical signal having a Y (or TM) polarization. The Y polarized modulated optical signal is also provided to PBC **414**, which combines the X and Y polarized modulated optical signals to provide a polarization multiplexed ("dual-pol") modulated optical signal onto optical fiber **416**, for example, which may be included as a segment of optical fiber in optical communication path **111**.

The polarization multiplexed optical signal output from D/A and optics block **401** includes subcarriers **SC0-SC7**, for example, such that each subcarrier has X and Y polarization components and I and Q components.

FIG. 5 shows an example of a transmitter (TX) DSP **402** in greater detail. TX DSP **402** may include FEC encoders **502-0** to **502-7**, each of which may receive a respective one of a plurality of data input **D0** to **D7**. FEC encoders **502-0** to **502-7** carry out forward error correction coding on a corresponding one of the switch outputs, such as, by adding parity bits to the received data. FEC encoders **502-0** to **502-7** may also provide timing skew between the subcarriers to correct for skew introduced during transmission over one or more optical fibers. In addition, FEC encoders **502-0** to **502-7** may interleave the received data.

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Each of FEC encoders **502-0** to **502-7** provides an output to a corresponding one of a plurality of bits-to-symbol circuits, **504-0** to **504-7** (collectively referred to herein as “**504**”). Each of bits to symbol circuits **504** may map the encoded bits to symbols on a complex plane. For example, bits to symbol circuits **504** may map four bits to a symbol in a dual-polarization quadrature phase shift keying (QPSK) constellation, or may map two bits to a symbol in a binary phase shift keying (BPSK) constellation. Each of bits to symbol circuits **504** provides first symbols, having the complex representation $XI+j*XQ$, associated with a respective one of the data input, such as **D0**, to DSP portion **503**. Data indicative of such first symbols may be carried by the X polarization component of each subcarrier **SC0-SC7**.

Each of bits to symbol circuits **504** may further provide second symbols having the complex representation $YI+j*YQ$, also associated with a corresponding one of data inputs **D0** to **D7**. Data indicative of such second symbols, however, is carried by the Y polarization component of each of subcarriers **SC-0** to **SC-7**.

As further shown in FIG. 5, each of the first symbols output from each of bits to symbol circuits **504** is supplied to a respective one of first overlap and save buffers **505-0** to **505-7** (collectively referred to herein as overlap and save buffers **505**) that may buffer 256 symbols, for example. Each of overlap and save buffers **505** may receive 128 of the first symbols or another number of such symbols at a time from a corresponding one of bits to symbol circuits **504**. Thus, overlap and save buffers **505** may combine 128 new symbols from bits to symbol circuits **505**, with the previous 128 symbols received from bits to symbol circuits **505**.

Each overlap and save buffer **505** supplies an output, which is in the time domain, to a corresponding one of fast Fourier Transform (FFT) circuits **506-0** to **506-7** (collectively referred to as “**FFTs 506**”). In one example, the output includes 256 symbols or another number of symbols. Each of **FFTs 506** converts the received symbols to the frequency domain using or based on, for example, a fast Fourier transform. Each of **FFTs 506** may include 256 memories or registers, also referred to as frequency bins or points, that store frequency components associated with the input symbols. Each of replicator components **507-0** to **507-7** may replicate the 256 frequency components associated with of **FFTs 506** and store such components in **512** or another number of frequency bins (e.g., for $T/2$ based filtering of the subcarrier) in a respective one of the plurality of replicator components. Such replication may increase the sample rate. In addition, replicator components or circuits **507-0** to **507-7** may arrange or align the contents of the frequency bins to fall within the bandwidths associated with pulse shaped filter circuits **508-0** to **508-7** described below.

Each of pulse shape filter circuits **508-0** to **508-7** may apply a pulse shaping filter to the data stored in the **512** frequency bins of a respective one of the plurality of replicator components **507-0** to **507-7** to thereby provide a respective one of a plurality of filtered outputs, which are multiplexed and subject to an inverse FFT, as described below. Pulse shape filter circuits **508-1** to **508-7** calculate the transitions between the symbols and the desired subcarrier spectrum so that the subcarriers can be spectrally packed together for transmission, e.g., with a close frequency separation. Pulse shape filter circuits **508-0** to **508-7** may also be used to introduce timing skew between the subcarriers to correct for timing skew induced by links between nodes shown in FIG. 1, for example. Multiplexer component **509**, which may include a multiplexer circuit or memory, may receive the filtered outputs from pulse shape filter circuits

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508-0 to **508-7**, and multiplex or combine such outputs together to form an element vector.

Next, IFFT circuit or component **510-1** may receive the element vector and provide a corresponding time domain signal or data based on an inverse fast Fourier transform (IFFT). In one example, the time domain signal may have a rate of 64 gigasamples per second (GSample/s). Take last buffer or memory circuit **511-1** may select the last **524** or another number of samples from an output of IFFT component or circuit **510-1** and supply the samples to DACs **404-1** and **404-2** at 64 GSample/s, for example. As noted above, DAC **404-1** is associated with the in-phase (I) component of the X pol signal and DAC **404-2** is associated with the quadrature (Q) component of the Y pol signal. Accordingly, consistent with the complex representation $XI+jXQ$, DAC **404-1** receives values associated with XI and DAC **404-2** receives values associated with jXQ . Based on these inputs, DACs **404-1** and **404-2** provide analog outputs to MZMD **406-1** and MZMD **406-2**, respectively, as discussed above.

As further shown in FIG. 5, each of bits to symbol circuits **504-0** to **504-7** outputs a corresponding one of symbols indicative of data carried by the Y polarization component of the polarization multiplexed modulated optical signal output on an optical fiber. As further noted above, these symbols may have the complex representation $YI+j*YQ$. Each such symbol may be processed by a respective one of overlap and save buffers **515-0** to **515-7**, a respective one of FFT circuits **516-0** to **516-7**, a respective one of replicator components or circuits **517-0** to **517-7**, pulse shape filter circuits **518-0** to **518-7**, multiplexer or memory **519**, IFFT **510-2**, and take last buffer or memory circuit **511-2**, to provide processed symbols having the representation $YI+j*YQ$ in a manner similar to or the same as that discussed above in generating processed symbols $XI+j*XQ$ output from take last circuit **511-1**. In addition, symbol components YI and YQ are provided to DACs **404-3** and **404-4**, respectively. Based on these inputs, DACs **404-3** and **404-4** provide analog outputs to MZMD **406-3** and MZMD **406-4**, respectively, as discussed above.

While FIG. 5 shows DSP **402** as including a particular quantity and arrangement of functional components, in some implementations, DSP **402** may include additional functional components, fewer functional components, different functional components, or differently arranged functional components. In addition, typically the number of overlap and save buffers, **FFTs**, replicator circuits, and pulse shape filters associated with the X component may be equal to the number of data inputs, and the number of such circuits associated with the Y component may also be equal to the number of switch outputs. However, in other examples, the number of data inputs may be different than the number of these circuits.

FIG. 6 illustrates an example of a receiver **246** of a primary transceiver **110-1** consistent with the present disclosure.

The receiver **246** in this example is configured to receive data from $M=8$ servers by a composite carrier signal that includes 8 optical carriers. As shown in FIG. 6, optical receiver **246** may include a receiver (RX) optics and A/D block **600**. Optical receiver **246** may also include a processor circuit, e.g., RX DSP **650**, which, in conjunction with the RX optics and A/D block **600**, may carry out coherent detection. Block **600** may include a polarization splitter **605** with first (**605-1**) and second (**605-2**) outputs, a frequency control circuit **607**, one of a plurality of local oscillator (LO) lasers **610**, 90 degree optical hybrids or mixers **620-1** and **620-2** (referred to generally as hybrid mixers **620** and

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individually as hybrid mixer **620**), detectors **630-1** and **630-2** (referred to generally as detectors **630** and individually as detector **630**, each including either a single photodiode or balanced photodiode), AC coupling capacitors **632-1** and **632-2**, transimpedance amplifiers/automatic gain control circuits TIA/AGC **634-1** and **634-2**, ADCs **640-1** and **640-2** (referred to generally as ADCs **640** and individually as ADC **640**), and an RX DSP **650**.

Polarization beam splitter (PBS) **605** may include a polarization splitter that receives an input polarization combined plurality of optical carriers supplied by optical fiber link **601**, which may be, for example, an optical fiber segment as part of one of an optical communication path. PBS **605** may split the incoming optical signal into the two X and Y orthogonal polarization components. The Y component may be supplied to a polarization rotator **606** that rotates the polarization of the Y component to have the X polarization. Hybrid mixers **620** may combine the X and rotated Y polarization components with light from local oscillator laser **610**. For example, hybrid mixer **620-1** may combine a first polarization signal (e.g., the component of the incoming optical signal having a first or X (TE) polarization output from PBS port **605-1**) with light from local oscillator **610**, and hybrid mixer **620-2** may combine the rotated polarization signal (e.g., the component of the incoming optical signal having a second or Y (TM) polarization output from PBS port **605-2**) with the light from local oscillator **610**. In one example, polarization rotator **690** may be provided at PBS output **605-2** to rotate Y component polarization to have the X polarization.

In some examples, the local oscillator **610** may be a semiconductor laser, which may be tuned thermally or through current adjustment. If thermally tuned, the temperature of the local oscillator laser **610** is controlled with a thin film heater, for example, provided adjacent the local oscillator laser. The local oscillator **610** may be an electrical local oscillator. The RX optics and A/D block **600** may include a frequency control circuit **607** for controlling the local oscillator **610**. The frequency control circuit **607** may be coupled to the local oscillator laser **610** to control the frequency of light output from the local oscillator laser **610**. The current supplied to the laser may be controlled, if the local oscillator laser is current tuned. The local oscillator laser **610** may be a semiconductor laser, such as a distributed feedback laser or a distributed Bragg reflector laser.

The RX optics and A/D block **600** may include a photodetector circuit that may include the detectors **630**, capacitors **632**, TIA/AGCs **634**, and ADCs **640**. The photodetector circuit may receive a portion of the light output from the local oscillator laser **610** and a portion of the combined plurality of optical carriers. The photodetector circuit may convert the plurality of optical carriers to electrical signals. Detectors **630** may detect mixing products output from the optical hybrids, to form corresponding voltage signals, which are subject to AC coupling by capacitors **632-1** and **632-2**, as well as amplification and gain control by TIA/AGCs **634-1** and **634-2**. The outputs of TIA/AGCs **634-1** and **634-2** and ADCs **640** may convert the voltage signals to digital samples. For example, two detectors or photodiodes **630-1** may detect the X polarization signals to form the corresponding voltage signals, and a corresponding two ADCs **640-1** may convert the voltage signals to digital samples for the first polarization signals after amplification, gain control and AC coupling. Similarly, two detectors **630-2** may detect the rotated Y polarization signals to form the corresponding voltage signals, and a corresponding two ADCs **640-2** may convert the voltage signals to digital

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samples for the second polarization signals after amplification, gain control and AC coupling.

The processor circuit, e.g., RX DSP **650**, may supply the plurality of data channels based on an output of the local oscillator **610** and the electrical signals from the photodetector circuit. For example, RX DSP **650** may process the electrical signals from the photodetector circuit and the electrical local oscillator output to supply data channels D0 to D7 to the Tier 1 switch **102**.

In some examples, the optics and A/D block **600** can include one of a plurality of optical filters **602**. The optical filter **602** may select a respective plurality of carriers from a group of carriers provided to the receiver **246**. For example, the optical filter **602** may isolate the plurality of carriers by allowing the portion of the group of carriers that is within a specified frequency range of the plurality of carriers to pass through the optical filter **602**, while reflecting the other portions of the group of carriers.

While FIG. 6 shows optical receiver **246** as including a particular quantity and arrangement of components, in some implementations, optical receiver **246** may include additional components, fewer components, different components, or differently arranged components. The quantity of detectors **630** and/or ADCs **640** may be selected to implement an optical receiver **302** that is capable of receiving a polarization multiplexed signal. In some instances, one of the components illustrated in FIG. 6 may carry out a function described herein as being carried out by another one of the components illustrated in FIG. 6.

FIG. 7 illustrates exemplary components of receiver digital signal processor (DSP) **650**. As noted above, analog-to-digital (A/D) circuits **640-1** and **640-2** output digital samples corresponding to the analog inputs supplied thereto. In one example, the samples may be supplied by each A/D circuit at a rate of 64 GSamples/s. The digital samples correspond to symbols carried by the X polarization of the optical subcarriers and may be represented by the complex number $XI + jXQ$. The digital samples may be provided to overlap and save buffer **705-1**, as shown in FIG. 7. FFT component or circuit **710-1** may receive the 2048 vector elements, for example, from the overlap and save buffer **705-1** and convert the vector elements to the frequency domain using, for example, a fast Fourier transform (FFT). The FFT component **710-1** may convert the 2048 vector elements to 2048 frequency components, each of which may be stored in a register or "bin" or other memory, as a result of carrying out the FFT.

The frequency components may then be demultiplexed, and groups of such components may be supplied to a respective one of chromatic dispersion equalizer circuits CDEQ **712-1-0** to **712-1-7**, each of which may include a finite impulse response (FIR) filter that corrects, offsets or reduces the effects of, or errors associated with chromatic dispersion of the transmitted optical subcarriers. Each of CDEQ circuits **712-1-0** to **712-1-7** supplies an output to a corresponding polarization mode dispersion (PMD) equalizer circuit **725-0** to **725-7**.

It is noted that digital samples output from A/D circuits **640-2** associated with Y polarization components of each subcarrier may be processed in a similar manner to that of digital samples output from A/D circuits **640-1** and associated with the X polarization component of each subcarrier. Namely, overlap and save buffer **705-2**, FFT **710-2** and CDEQ circuits **712-2-0** to **712-2-7** may have a similar structure and operate in a similar fashion as buffer **705-1**, FFT **710-1** and CDEQ circuits **712-1-0** to **712-1-7**, respectively. For example, each of CDEQ circuits **712-2-0** to **712-2-7**

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may include an FIR filter that corrects, offsets, or reduces the effects of, or errors associated with chromatic dispersion of the transmitted optical subcarriers. In addition, each of CDEQ circuits **712-2-0** to **712-2-7** provide an output to a corresponding one of PMDEQ **725-0** to **725-7**.

As further shown in FIG. 7, the output of one of the CDEQ circuits, such as CDEQ **712-1-0** may be supplied to clock phase detector circuit **713** to determine a clock phase or clock timing associated with the received subcarriers. Such phase or timing information or data may be supplied to ADCs **640-1** and **640-2** to adjust or control the timing of the digital samples output from ADCs **640-1** and **640-2**.

Each of PMDEQ circuits **725** may include another FIR filter that corrects, offsets or reduces the effects of, or errors associated with PMD of the transmitted optical subcarriers. Each of PMDEQ circuits may supply **725** supplies a first output to a respective one of IFFT components or circuits **730-0-1** to **730-7-1** and a second output to a respective one of IFFT components or circuits **730-0-2** to **730-7-2**, each of which may convert a 256 element vector, in this example, back to the time domain as 256 samples in accordance with, for example, an inverse fast Fourier transform (IFFT).

Time domain signals or data output from IFFT **730-0-1** to **730-7-1** are supplied to a corresponding one of Xpol carrier phase correction circuits **740-1-1** to **740-7-1**, which may apply carrier recovery techniques to compensate for X polarization transmitter (e.g., laser **408**) and receiver (e.g., local oscillator laser **610**) linewidths. In some implementations, each carrier phase correction circuit **740-1-1** to **740-7-1** may compensate or correct for frequency and/or phase differences between the X polarization of the transmit signal and the X polarization of light from the local oscillator **600** based on an output of Xpol carrier recovery circuit **740-0-1**, which performs carrier recovery in connection with one of the subcarrier based on the outputs of IFFT **730-0-1**. After such X polarization carrier phase correction, the data associated with the X polarization component may be represented as symbols having the complex representation $XI+j*YQ$ in a constellation, such as a QPSK constellation, a BPSK constellation, or a constellation associated with another modulation formation, such as an m-quadrature amplitude modulation (QAM), m being an integer. In some implementations, the taps of the FIR filter included in one or more of PMDEQ circuits **725** may be updated based on the output of at least one of carrier phase correction circuits **740-0-1** to **740-7-01**.

In a similar manner, time domain signals or data output from IFFT **730-0-2** to **730-7-2** are supplied to a corresponding one of Ypol carrier phase correction circuits **740-0-2** to **740-7-2**, which may compensate or correct for Y polarization transmitter (e.g., laser **408**) and receiver (e.g., local oscillator laser **610**) linewidths. In some implementations, each carrier phase correction circuit **740-0-2** to **740-7-2** may also corrector or compensate or correct for frequency and/or phase differences between the Y polarization of the transmit signal and the Y polarization of light from the local oscillator **610**. After such Y polarization carrier phase correction, the data associated with the Y polarization component may be represented as symbols having the complex representation $YI+j*YQ$ in a constellation, such as a QPSK constellation, a BPSK constellation, or a constellation associated with another modulation formation, such as an m-quadrature amplitude modulation (QAM), m being an integer. In some implementations, the output of one of circuits **740-0-2** to **740-7-2** may be used to update the taps of the FIR filter included in one or more of PMDEQ circuits **725** instead of

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or in addition to the output of at least one of the carrier recovery circuits **740-0-1** to **740-7-1**.

As further shown in FIG. 7, the output of carrier recovery circuits, e.g., carrier recovery circuit **740-0-1**, may also be supplied to carrier phase correction circuits **740-1-1** to **740-7-1** and **740-0-2** to **740-7-2** whereby the phase correction circuits may determine or calculate a corrected carrier phase associated with each of the received subcarriers based on one of the recovered carriers, instead of providing multiple carrier recovery circuits, each of which being associated with a corresponding subcarrier.

Each of the symbols to bits circuits or components **745-0-1** to **745-7-1** may receive the symbols output from a corresponding one of circuits **740-0-1** to **740-7-1** and map the symbols back to bits. For example, each of the symbol to bits components **745-0-1** to **745-7-1** may map one X polarization symbol, in a QPSK or m-QAM constellation, to Z bits, where Z is an integer. For dual-polarization QPSK modulated subcarriers, Z is four. Bits output from each of component **745-0-1** to **745-7-1** are provided to a corresponding one of FEC decoder circuits **760-0** to **760-7**.

Y polarization symbols are output form a respective one of circuits **740-0-2** to **740-7-2**, each of which having the complex representation $YI+j*YQ$ associated with data carried by the Y polarization component. Each Y polarization, like the X polarization symbols noted above, may be provided to symbols to a corresponding one of bit to symbol circuits or components **745-0-2** to **745-7-2**, each of which having a similar structure and operating a similar manner as symbols to bits component **745-0-1** to **745-7-1**. Each of circuits **745-0-2** to **745-7-2** may provide an output to a corresponding one of FEC decoder circuits **760-0** to **760-7**.

Each of FEC decoder circuits **760** may remove errors in the outputs of symbol to bit circuits **745** using forward error correction. Such error corrected bits may be supplied as a corresponding one of outputs D0 to D7.

While FIG. 7 shows DSP **650** as including a particular quantity and arrangement of functional components, in some implementations, DSP **650** may include additional functional components, fewer functional components, different functional components, or differently arranged functional components.

FIG. 8 illustrates an example transmitter **230** of the secondary transceiver **116-1A** consistent with the present disclosure. The transmitter **230** may include a digital signal processor (DSP) **802** and a D/A and optics block **801**. The construction and operation of the DSP **802** is similar to the construction and operation of the DSP **402**. The construction and operation of the D/A and optics block **801** is similar to the construction and operation of the D/A and optics block **401**. In this example, the DSP **802** receives one of a plurality of data channels from a server. Based on the one of the plurality of data channels, DSP **802** may supply an output to D/A and optics block **801**, which converts the digital signal received from DSP **802** into a corresponding one of a plurality of analog RF signals. D/A and optics block **801** may also adjust the voltages or other characteristics of the analog RF signals to provide drive signals to a corresponding optical modulator. The D/A and optics block **801** may output a carrier signal with data modulated onto a single carrier. The D/A and optics block **801** may output the carrier signal to the optical platform for aggregation with additional carrier signals and delivery to upstream components of the optical network.

FIG. 9 illustrates an example of a receiver **224** of a secondary transceiver **116-1A**. The receiver **224** may include an optics and A/D block **900**. Optical receiver **224**

may also include a processor circuit, e.g., RX DSP **950**, which, in conjunction with the RX optics and A/D block **900**, may carry out coherent detection. The construction and operation of the DSP **950** is similar to the construction and operation of the DSP **650**. The construction and operation of the optics and A/D block **900** is similar to the construction and operation of the optics and A/D block **600**. In this example, the receiver **224** receives an output including subcarriers SC0 to SC7. For example, the receiver **224** may receive the output from the optical platform **112-1**.

In order to demodulate subcarriers SC0 to SC7, the local oscillator of the optics and A/D block **900** may be tuned to output light having a wavelength or frequency relatively close to one or more of the desired subcarrier wavelengths or frequencies to thereby cause a beating between the local oscillator light and the subcarriers. The local oscillator may also be, for example, an electrical local oscillator. The RX optics and A/D block **900** may include a frequency control circuit, e.g., the frequency control circuit **607**. The frequency control circuit **607** may be coupled to the local oscillator laser, e.g., the local oscillator **610**, to control the frequency of light output from the local oscillator laser **610**. The local oscillator may be tuned to output light having a wavelength or frequency that corresponds to a specific subcarrier with a corresponding server. For example, subcarrier SC0 may be associated with server **118-1A**. Thus, the local oscillator of the receiver **224** may be tuned to output light having a frequency close to the subcarrier frequency f_0 of subcarrier SC0.

The RX optics and A/D block **900** may include a photodetector circuit that may include the detectors, capacitors, TIA/AGCs, and ADCs. For example, the photodetector circuit may include the detectors **630**, capacitors **632**, TIA/AGCs **634**, and ADCs **640**. The photodetector circuit may receive a portion of the light output from the local oscillator laser **610** and a portion of the plurality of outputs of the optical platform **112-1**. The processor circuit, e.g., RX DSP **950**, may supply one of the plurality of data channels based on a frequency of the light output from the local oscillator. For example, The RX DSP **950** may supply D0 to the server **118-1A** based on the frequency f_0 output from the local oscillator.

In some examples, the optics and A/D block can include one of a plurality of optical filters, e.g., the optical filter **602**. The optical filter **602** may be, for example, a Fabry-Pérot interferometer (FPI). The optical filter **602** may select a respective one of the plurality of optical subcarriers. For example, the optical filter **602** may isolate a subcarrier by allowing the portion of the output that is within a specified frequency range of the subcarrier to pass through the optical filter **602**, while reflecting the other portions of the output. For example, if SC0 is associated with the server **118-1A**, the optical filter **602** may select the frequencies within the frequency range of SC0, and allow those frequencies to pass through the optical filter **602**, while reflecting the frequencies outside of the frequency range of SC0.

In some implementations, the receiver **224** may isolate the subcarrier channel associated with the server using an electronic filter circuit. An electronic filter circuit may be a FIR filter, e.g., a FIR filter within each of CDEQ circuits **712-2-0** to **712-7**. The FIR filter may correct, offset, or reduces the effects of, or errors associated with chromatic dispersion of the transmitted optical subcarriers. The FIR filter may also be configured to allow the portion of the output within the appropriate subcarrier frequency range to pass, while attenuating the portions of the output that are not within the

appropriate subcarrier frequency ranges. The RX DSP **950** may then supply the corresponding data channels to the server.

FIG. **10** is a flow diagram that illustrates an example of a process **1000** for transmitting an optical signal downstream from a Tier 1 switch to a server.

Generally, the process **1000** includes supplying a plurality of data channels from a Tier 1 switch (**1002**), receiving, by an optical modulator, a plurality of RF signals associated with the plurality of data channels (**1004**), supplying, from the optical modulator, a plurality of optical subcarriers based on the plurality of data channels (**1006**), receiving, by an optical platform, the plurality of optical subcarriers (**1008**), supplying, from the optical platform, at least one of the plurality of subcarriers (**1010**), receiving, by a plurality of receivers, one or more of the plurality of optical subcarriers (**1012**), supplying, from the plurality of receivers, one or more of the plurality of data channels to one of a plurality of servers (**1014**), and receiving, by a plurality of servers, one or more of the plurality of data channels (**1016**).

During **1002**, a Tier 1 switch supplies a plurality of data channels to a primary transceiver. The Tier 1 switch can be, for example, the Tier 1 switch **102**. The primary transceiver can be, for example, primary transceiver **110-1**. The data channels can be, for example, the data channels D1 to DM. The primary transceiver may receive the data channels from the Tier 1 switch through an I/O port.

During **1004**, an optical modulator receives a plurality of RF signals associated with the plurality of data channels. The optical modulator modulates the RF signals associated with the plurality of data channels onto one or more subcarriers. Each of the subcarriers may be associated with a specific server.

During **1006**, the optical modulator supplies a plurality of optical subcarriers based on the plurality of data channels. The optical subcarriers can be, for example, the subcarriers SC1 to SCM. The number M of subcarriers may correspond to the number of destination servers. The bandwidth of each subcarrier may be $1/M$ times the bandwidth of the plurality of optical subcarriers.

During **1008**, an optical platform receives the plurality of optical subcarriers. The optical platform can be, for example, the optical platform **112-1**. The optical platform may receive the plurality of optical subcarriers via one or more optical fibers.

During **1010**, the optical platform supplies at least one of the plurality of optical subcarriers. The optical platform may supply the at least one of the plurality of optical subcarriers using an optical splitter or demultiplexer, for example, the optical splitter **114-1**.

During **1012**, a plurality of receivers receives one or more of the plurality of optical subcarriers. The one or more of the plurality of optical subcarriers may be, for example, the outputs **120-1A** to **120-1M**. The plurality of receivers can be, for example, the receivers **224** of the secondary transceivers **116-1A** to **116-1M**. The plurality of receivers can receive the plurality of optical subcarriers through one or more optical fibers.

During **1014**, the plurality of receivers supplies one or more of the plurality of data channels to one of a plurality of servers. The plurality of receivers may supply, from the plurality of data channels, the data channels associated with a server.

During **1016**, the plurality of servers receives one or more of the plurality of data channels. The one or more of the plurality of data channels can be, for example, the data

channel D1. The plurality of servers may be, for example, the servers **118-1A** to **118-1M**.

FIG. **11** is a flow diagram that illustrates an example of a process **1100** for transmitting an optical signal upstream from a server to Tier 1 switch.

Generally, the process **1100** includes receiving a data channel from a server (**1102**), generating one of a plurality of optical carriers (**1104**), transmitting the one of the plurality of optical carriers to an optical platform (**1106**), combining the plurality of optical carriers onto an optical fiber (**1108**), receiving the combined plurality of optical carriers (**1110**), and, based on the plurality of optical carriers, supplying the plurality of data channels to a Tier 1 switch (**1112**).

During **1102**, a secondary transceiver receives a data channel from a server. The server can be, for example, the server **118-1A**. The secondary transceiver can be, for example, the secondary transceiver **116-1A**. The data channel can be, for example, the data channel D1'.

During **1104**, the secondary transceiver generates one of a plurality of optical carriers from the data channel. The one of a plurality of optical carriers can be, for example, one of the carrier signals **122-1A** to **122-1M**.

During **1106**, the secondary transceiver transmits the one of a plurality of optical carriers to an optical platform. The optical platform can be, for example, the optical platform **112-1**. The secondary transceiver can transmit the one of a plurality of optical carriers to the optical platform via one or more optical fibers.

During **1108**, the optical platform combines the plurality of optical carriers onto an optical fiber. The optical platform can combine the plurality of optical carriers using, for example, the optical power combiner or multiplexer **115-1**. The combined plurality of optical carriers can be, for example, the combined plurality of optical carriers **124-1**.

During **1110**, the primary transceiver receives the combined plurality of optical carriers. The primary transceiver demodulates the optical carriers, from the composite carrier signal.

During **1114**, the primary transceiver outputs the data channels to a Tier 1 switch. The Tier 1 switch can be, for example, the Tier 1 switch **102**. The data channels can be, for example, the data channels D1' to DM'. The primary transceiver can output the data channels to the Tier 1 switch through an I/O port.

Embodiments of the disclosure and all of the functional operations described in this specification may be implemented in digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. Embodiments of the disclosure may be implemented as one or more computer program products, i.e., one or more modules of computer program instructions encoded on a computer-readable medium for execution by, or to control the operation of, data processing apparatus. The computer readable medium may be a non-transitory computer readable storage medium, a machine-readable storage device, a machine-readable storage substrate, a memory device, a composition of matter effecting a machine-readable propagated signal, or a combination of one or more of them. The term "data processing apparatus" encompasses all apparatus, devices, and machines for processing data, including by way of example a programmable processor, a computer, or multiple processors or computers. The apparatus may include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes

processor firmware, a protocol stack, a database management system, an operating system, or a combination of one or more of them. A propagated signal is an artificially generated signal, e.g., a machine-generated electrical, optical, or electromagnetic signal that is generated to encode information for transmission to suitable receiver apparatus.

A computer program (also known as a program, software, software application, script, or code) may be written in any form of programming language, including compiled or interpreted languages, and it may be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program does not necessarily correspond to a file in a file system. A program may be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program may be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

The processes and logic flows described in this specification may be performed by one or more programmable processors executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows may also be performed by, and apparatus may also be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit).

Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Moreover, a computer may be embedded in another device, e.g., a tablet computer, a mobile telephone, a personal digital assistant (PDA), a mobile audio player, a Global Positioning System (GPS) receiver, to name just a few. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media, and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory may be supplemented by, or incorporated in, special purpose logic circuitry.

To provide for interaction with a user, embodiments of the disclosure may be implemented on a computer having a display device, e.g., a CRT (cathode ray tube) or LCD (liquid crystal display) monitor, for displaying information to the user and a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user may provide input to the computer. Other kinds of devices may be used to provide for interaction with a user as well; for example, feedback provided to the user may be any form of sensory

feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user may be received in any form, including acoustic, speech, or tactile input.

Embodiments of the disclosure may be implemented in a computing system that includes a back end component, e.g., as a data server, or that includes a middleware component, e.g., an application server, or that includes a front end component, e.g., a client computer having a graphical user interface or a Web browser through which a user may interact with an implementation of the disclosure, or any combination of one or more such back end, middleware, or front end components. The components of the system may be interconnected by any form or medium of data channels communication, e.g., a communication network. Examples of communication networks include a local area network ("LAN") and a wide area network ("WAN"), e.g., the Internet.

The computing system may include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

Although a few implementations have been described in detail above, other modifications are possible. For example, while a client application is described as accessing the delegate(s), in other implementations the delegate(s) may be employed by other applications implemented by one or more processors, such as an application executing on one or more servers. In addition, the logic flows depicted in the figures do not require the particular order shown, or sequential order, to achieve desirable results. In addition, other actions may be provided, or actions may be eliminated, from the described flows, and other components may be added to, or removed from, the described systems. Accordingly, other implementations are within the scope of the following claims.

While this specification contains many specific implementation details, these should not be construed as limitations on the scope of any disclosure or of what may be claimed, but rather as descriptions of features that may be specific to particular embodiments of particular disclosures. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system modules and components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

Particular embodiments of the subject matter have been described. Other embodiments are within the scope of the following claims. For example, the actions recited in the claims can be performed in a different order and still achieve desirable results. As one example, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In certain implementations, multitasking and parallel processing may be advantageous.

What is claimed is:

1. A system, comprising:
 - a Tier 1 switch that supplies a plurality of data channels; a first transceiver including a transmitter that receives the plurality of data channels, the transmitter including:
 - a laser, and
 - an optical modulator that receives a plurality of radio frequency (RF) signals associated with the plurality of data channels, such that the optical modulator supplies a plurality of optical subcarriers by modulating an output of the laser, each of which being associated with a respective one of the plurality of data channels;
 - an optical platform, including one of an optical demultiplexer or a splitter, that receives the plurality of optical subcarriers, the optical platform having a plurality of outputs, each of which supplying at least one of the plurality of subcarriers;
 - a second transceiver that includes a first receiver coupled to a first output of the optical platform, the first receiver being operable to receive the plurality of optical subcarriers and supply a group of data channels of the plurality data channels based on a group of the plurality of optical subcarriers;
 - a third transceiver that includes a second receiver coupled to a second output of the optical platform, the second receiver being operable to receive the plurality of optical subcarriers and supply one data channel of the plurality of data channels based on one of the plurality of optical subcarriers; and
 - a plurality of servers, a first one of the plurality of servers being operable to receive at least one of the group of data channels from the first receiver and a second one of the plurality of servers being operable to receive said one of the plurality of data channels from the second receiver.
2. A system in accordance with claim 1, wherein the optical platform includes an optical splitter that supplies a copy of the plurality of subcarriers at each of the plurality of outputs.
3. A system in accordance with claim 1, wherein each of the plurality of optical subcarriers is a Nyquist subcarrier.
4. A system in accordance with claim 1, wherein each of the plurality of data channels supplied to the transmitter constitutes a respective one of a plurality of electrical signals.
5. A system in accordance with claim 1, wherein the Tier 1 switch includes a housing, the housing including the first transceiver.
6. A system in accordance with claim 1, wherein each of plurality of optical subcarriers is modulated in accordance with a modulation format, the modulation format being selected from an m-quadrature amplitude modulation (QAM), m being an integer, quadrature phase shift keying (QPSK), and binary phase shift keying (BPSK).
7. A system in accordance with claim 1, wherein the first receiver includes a first local oscillator laser and the second receiver includes a second local oscillator laser.

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8. A system in accordance with claim 1, wherein the second receiver includes:

- a local oscillator laser;
- a control circuit, the control circuit being coupled to the local oscillator laser to thereby control a frequency of light output from the local oscillator laser;
- a photodetector circuit configured to receive at least a portion of the light output from the local oscillator laser and at least part of one of the plurality of outputs of the optical platform; and

a processor circuit that supplies said one of the plurality of data channels based on an output of the photodetector circuit.

9. A system in accordance with claim 8, wherein the processor supplies said one of the plurality of data channels further based on a frequency of the light output from the local oscillator.

10. A system in accordance with claim 9, wherein the processor circuit includes a filter circuit, the processor circuit supplying said one of the plurality of data channels based on an output of the filter circuit.

11. A system in accordance with claim 1, second receiver includes a filter operable to select said one of the plurality of optical subcarriers.

12. A system, comprising:

- a plurality of servers, each of which supplying a corresponding one of a plurality of data channels;
- a plurality of first transceivers, each of which including a respective one of a plurality of transmitters, each of which receiving a corresponding one of the plurality of data channels, each of the plurality of transmitters providing a respective one of a plurality of optical carriers, each of the plurality of optical carriers being modulated in accordance with a respective one of a plurality of radio frequency (RF) signals, each RF signal being based on a respective one of the plurality of data channels;

an optical platform operable to combine the plurality of optical carriers into a composite carrier signal having a

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center frequency, the optical platform also being operable to supply the composite carrier signal onto an optical fiber;

- a second transceiver including a receiver that receives the composite carrier signal, the receiver including a local oscillator, such that based on the composite optical signal and an output of the local oscillator, the receiver supplies the plurality of data channels; and
- a Tier 1 switch that receives the plurality of data channels.

13. A system in accordance with claim 12, wherein the optical platform includes an optical power combiner.

14. A system in accordance with claim 12, wherein the optical platform includes an optical multiplexer.

15. A system in accordance with claim 12, wherein the optical platform includes an arrayed waveguide grating.

16. A system in accordance with claim 12, wherein the local oscillator is operable to supply a local oscillator signal, the receiver including:

- an optical hybrid circuit operable to supply a plurality of mixing products based on the local oscillator signal and the composite carrier signal;

a photodetector circuit operable to provide electrical signals based on the plurality of mixing products.

17. A system in accordance with claim 12, wherein each of the plurality of optical carriers is a Nyquist carrier.

18. A system in accordance with claim 12, wherein each of the plurality of data channels constitutes a respective one of a plurality of electrical signals.

19. A system in accordance with claim 12, wherein the Tier 1 switch includes a housing, the housing including the second transceiver.

20. A system in accordance with claim 12, wherein each of plurality of optical carriers is modulated in accordance with a modulation format, the modulation format being selected from an m-quadrature amplitude modulation (QAM), m being an integer, quadrature phase shift keying (QPSK), and binary phase shift keying (BPSK).

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